MATH 475: Honours Partial Differential Equations

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Abstract

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1 Introduction

Textbook used is Rustum Choksi, *Partial Differential Equations, A First Course*, American Mathematical Society. Course grading is divided as follows:

- 10% 5-6 assignments.
- 10% attendance and participation.
- 10% 3-5 quizzes.
- 20% midterm.
- 50% final exam.

2 Background

3 Characteristics

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3.1 Partial Differential Equations

Definition 1 (Partial Differential Equation)**.** *A partial differential equation (PDE) is a differential equation whose unknown function depends on more than one independent variable.*

Remark 1. *Physics rarely hands us a function u directly. It gives a law for how u changes. Example: heat flows from hot to cold, so the temperature $u(x,t)$ in a room changes in time wherever it varies in space. Recovering u from such a relation is the whole subject.*

3.2 Classification

3.2.1 Order

Definition 2 (Order)**.** *The order of a PDE is the order of the highest derivative appearing in it.*

Example. *$u_{xx} = 0$ is second order, and $u_t + uu_x = 0$ is first order.*

3.2.2 Linearity

Definition 3 (Linear PDE)**.** *A PDE is linear if it can be written as $L(u) = f(x)$, where f does not involve u and L satisfies*

$$L(u_1 + c\,u_2) = L(u_1) + c\,L(u_2). \tag{1}$$

Remark 2 (Practical test)**.** *Every term containing u or a derivative of u must satisfy:*

- *u and its derivatives appear only to the first power (no u^2, $(u_x)^2$),*

- *they are never multiplied together (no $u\,u_x$, $u_x u_y$),*
- *they never sit inside a function (no $\sin u$, e^u, $\ln u$),*
- *their coefficients depend only on the independent variables, never on u.*

Each violation creates cross terms, so the superposition identity above fails.

Example. *$u_{xx} + xy\,u_y + e^x u = \cos x$ is linear. Nonlinear: $u\,u_x + u_y = 0$ (product), $u_{xx} + (u_y)^2 = 0$ (power), $u_{xx} + \sin u = 0$ (inside a function).*

Definition 4 (Degrees of nonlinearity)**.** *Let k be the order of the PDE.*

- ***Semilinear:*** *linear in the order-k derivatives, with coefficients depending only on x; nonlinearity only in lower-order terms. Example: $u_{xx} + \sin u = 0$.*
- ***Quasilinear:*** *linear in the order-k derivatives, but their coefficients may depend on u and its lower-order derivatives. Example: Burgers, $u_t + u\,u_x = 0$.*
- ***Fully nonlinear:*** *the order-k derivatives themselves appear nonlinearly. Example: $(u_x)^2 + (u_y)^2 = 1$.*

3.2.3 Homogeneity

Definition 5 (Homogeneous)**.** *A linear PDE $L(u) = f(x)$ is homogeneous if $f \equiv 0$, and inhomogeneous otherwise.*

Example. *$u_t + au_x = x^3$ is linear with $L(u) = u_t + au_x$ and $f(x) = x^3$, so it is inhomogeneous.*

Example. *$u_t + uu_x = f(x)$ is nonlinear, because the term uu_x is not linear in u.*

Problem 1 (A1 Q2, Choksi 1.1)**.** *Classify the PDEs of Choksi 1.1 parts (b), (d), (e), (i), (j), (o) by order and as linear, semilinear, quasilinear or fully nonlinear. Then research the context of each: what it models, what u and the parameters mean, and the natural domain and auxiliary conditions.* Solution: p. 47.

3.3 Solutions

Definition 6 (Classical solution)**.** *Let the PDE be written as $F(u) = 0$ and let k be its order. A function u is a (classical) solution if*

(1) $u \in C^k$, and

(2) $F(u) = 0$ at every point of the domain.

Example (Transport equation)**.** *Let $u = u(x,t)$ and $F(u_t, u_x) = u_t + cu_x = 0$ with $c \in \mathbb{R}$. Let Φ be any differentiable function and set $u(x,t) = \Phi(x - ct)$. Then*

$$u_x = \Phi'(x - ct), \qquad u_t = -c\,\Phi'(x - ct), \tag{2}$$

so $u_t + cu_x = 0$. The whole family Φ solves the equation. Meaning: the initial profile Φ slides rigidly to the right at speed c.

Example (Heat equation)**.** *Let $u_t = c^2 u_{xx}$ and take the ansatz $u(x,t) = e^{-c^2k^2t}\sin(kx)$ for an arbitrary constant k. Then*

$$u_t = -c^2k^2u, \qquad u_x = k\,e^{-c^2k^2t}\cos(kx), \qquad u_{xx} = -k^2e^{-c^2k^2t}\sin(kx) = -k^2u, \tag{3}$$

so $u_t = c^2u_{xx}$. Meaning: the sine keeps its shape and decays, and high frequencies (large k) decay faster, at rate c^2k^2. Diffusion smooths.

Example (Wave equation)**.** *Let $u_{tt} = c^2u_{xx}$ and take $u(x,t) = \cos(kct)\sin(kx)$. Then*

$$u_{xx} = -k^2u, \qquad u_{tt} = -c^2k^2u, \tag{4}$$

so $u_{tt} = c^2u_{xx}$. Meaning: a standing wave that oscillates forever without decaying. Same spatial profile as the heat example, completely different time behaviour.

3.4 Principle of Superposition

Principle 1 (Superposition)**.** *Let L be linear.*

(1) If $L(u_1) = L(u_2) = 0$, then $L(u_1 + c\,u_2) = 0$ for every $c \in \mathbb{R}$. Solutions of a homogeneous linear PDE form a vector space.

(2) If $L(u_p) = f$ and $L(u_h) = 0$, then $L(u_p + u_h) = f$: one particular solution plus any homogeneous solution solves the inhomogeneous equation.

Proof. By linearity of L,

$$L(u_1{+}cu_2) = L(u_1){+}c\,L(u_2) = 0{+}c{\cdot}0 = 0, \qquad L(u_p{+}u_h) = f{+}0 = f. \ \square \tag{5}$$

Example. *For $u_t = c^2u_{xx}$, both*

$$u_1(x,t) = e^{-c^2k^2t}\sin(kx) \qquad \text{and} \qquad u_2(x,t) = e^{-c^2t}\sin(x) \tag{6}$$

are solutions, so $u(x,t) = a_1u_1(x,t) + a_2u_2(x,t)$ is a solution for any $a_1, a_2 \in \mathbb{R}$. This is how we will later build series solutions.

[1]

[1] 80% chance that what we have written so far is not on the final.

4 General Solutions

4.1 Solving by Direct Integration

Example. *Solve $u_x = 0$ for $u(x, y)$ on $\mathbb{R}^2$. The solution is $u(x, y) = f(y)$ for an arbitrary f.*

Example. *Solve $u_{xx} = 0$. Set $v(x, y) = u_x$. Then $v_x = 0$, so by the previous example $v(x, y) = f(y)$, hence $u_x = f(y)$. Fixing y and integrating in x (the "constant" of integration only has to be constant in x) gives*

$$u(x, y) = f(y)\, x + g(y). \tag{7}$$

Check: $u_x = f(y)$ and $u_{xx} = 0$.

Example. *Solve $u_{xy} = 0$. Set $v(x, y) = u_x$. Then $v_y = 0$, so $v(x, y) = f(x)$, hence $u_x = f(x)$. Fixing y and integrating in x gives*

$$u(x, y) = F(x) + g(y), \qquad F' = f. \tag{8}$$

Check: $u_x = F'(x)$ depends on x only, so $u_{xy} = 0$.

Remark 3. *In each case the general solution carries arbitrary* functions *rather than arbitrary constants. An ODE of order k has k free constants; a PDE has free functions of the remaining variables. Second order in one variable, or first order in each of two, gives two free functions.*

Problem 2 (A1 Q2, Choksi 1.2)**.** *Find the general solutions of $u_x = 0$, $u_{xy} = 0$ and $u_{xyz} = 0$ for $u = u(x, y, z)$.* Solution: p. 48.

4.2 Auxiliary Conditions

Auxiliary conditions (initial or boundary conditions) prescribe u, and possibly its derivatives, on a lower dimensional set. They cut the family of general solutions down to one. A typical problem on a domain $\Omega \subset \mathbb{R}^d$ reads

$$\Delta u = f(x) \quad \text{in } \Omega, \qquad u(x) = g(x) \quad \text{on } \partial\Omega. \tag{9}$$

The big three problems of the course:

Heat IVP:	$u_t = \alpha u_{xx},\ u(x, 0) = g(x),$	(10)
Wave IVP:	$u_{tt} = c^2 u_{xx},\ u(x, 0) = g(x),\ u_t(x, 0) = h(x),$	(11)
Dirichlet problem:	$\Delta u = 0$ in the unit disk $D,\ u = g$ on $\partial D.$	(12)

Definition 7 (Well-posed, Hadamard)**.** *A PDE with auxiliary conditions is well-posed if*

(1) ***Existence:*** *a solution exists,*

(2) ***Uniqueness:*** *there is only one,*

(3) ***Stability:*** *small changes in the data give small changes in the solution.*

Stability matters because data comes from measurements (with errors) and computers discretize.

Remark 4 (Counting conditions)**.** *For ODEs, number of conditions = order. For PDEs this fails: all three problems above are second order, but the wave IVP needs two conditions while heat and Laplace need one.*

Problem 3 (A1 Q2, Choksi 1.10)**.** *Show that the initial value problem for $u_t = u_{xxxx}$ is not well-posed.* Solution: p. 49.

4.3 Operator Viewpoint

Write a linear PDE as

$$Lu = f, \tag{13}$$

where L is the operator, u is the unknown and f is the input. The linear algebra analogue is

$$Mx = b, \qquad x, b \in \mathbb{R}^d, \qquad x = M^{-1}b, \tag{14}$$

so solving a linear PDE is the infinite-dimensional version of inverting M. Well-posedness already shows up here:

- M invertible: exactly one solution for every b.
- M not invertible, b in the range of M: solutions exist but are not unique (add anything in the null space).
- M not invertible, b not in the range: no solution.

Example. $M = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. *For $b = (2, 2)$ every $x = (2 - t, t)$ works: existence without uniqueness. For $b = (1, 2)$ there is no solution: the data is incompatible with the equation.*

Problem 4 (A1 Q2, Choksi 1.6)**.** *Well-posedness for $A\vec{x} = \vec{b}$, with A an $N \times N$ matrix.*

(a) What do existence and uniqueness mean, and when do they hold?

(b) What does stability mean? Assuming A is symmetric positive definite, relate it to the smallest eigenvalue of A.

Solution: p. 48.

4.4 How We Will Solve PDEs

(1) Assume a smooth solution exists.

(2) Work out what the equation and conditions force on it.

(3) In good cases this gives an explicit formula.

(4) Verify the formula really solves the problem. The verification is what makes it a proof.

5 Potential Theory

5.1 Newton's Problem

Place a point mass M at the origin and a test mass m at $x \in \mathbb{R}^3 \setminus \{0\}$. The gravitational force on m has magnitude $GmM/\|x\|^2$ and points back toward the origin:

$$\vec{F}(x) = -\frac{GmM}{\|x\|^2} \cdot \frac{x}{\|x\|} = -\frac{GmM}{\|x\|^3}\, x. \tag{15}$$

5.2 Toolkit

Lemma 1 (Derivative of distance is direction)**.** *Let $r(x) = \|x\|$ with $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Then for $x \neq 0$,*

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}, \qquad \nabla r(x) = \frac{x}{r} = \hat{x}. \tag{16}$$

Proof. Start from $r^2 = x_1^2 + x_2^2 + x_3^2$ and differentiate both sides with respect to x_i:

$$2r\frac{\partial r}{\partial x_i} = 2x_i \quad \Longrightarrow \quad \frac{\partial r}{\partial x_i} = \frac{x_i}{r}. \tag{17}$$

This requires $r \neq 0$, so the identity fails at the origin. □

Remark 5. *∇r is the unit vector pointing radially outward, so $\|\nabla r\| = 1$ everywhere it is defined.*

5.3 Radial Chain Rule

Proposition 1. *Let $f : (0, \infty) \to \mathbb{R}$ be differentiable and set $g(x) = f(r(x)) = f(\|x\|)$. Then for $x \neq 0$,*

$$\nabla g(x) = f'(r)\, \nabla r(x) = f'(r)\, \frac{x}{r}. \tag{18}$$

Proof. Componentwise, by the chain rule,

$$\frac{\partial g}{\partial x_i} = f'(r(x))\frac{\partial r}{\partial x_i} = f'(r)\frac{x_i}{r}. \tag{19}$$

□

So the gradient of any radial function points radially, and f' alone controls the magnitude.

Example (Gradient refresher, non-radial)**.** *For $g(x) = x_1 x_2 + x_3^2$,*

$$\nabla g(x) = \left(\frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2}, \frac{\partial g}{\partial x_3}\right) = (x_2,\ x_1,\ 2x_3). \tag{20}$$

Problem 5 (A1 Q1, vector calculus review). *(1) For $f(\vec{x}) = \|\vec{x}\|^2$ on $\mathbb{R}^3$, compute ∇f, the directional derivative at $\vec{x} = (1,0,0)$ in the direction $\vec{v} = \frac{1}{\sqrt{2}}(1,1,0)$, and the level sets. Sketch the plane version: contours and gradient arrows.*

(2) Show from the linear approximation that if $\nabla f(\vec{x}) \neq \vec{0}$, then among all unit vectors $\vec{v}$ the directional derivative $D_{\vec{v}} f(\vec{x})$ is maximized uniquely by $\vec{v} = \nabla f(\vec{x})/\|\nabla f(\vec{x})\|$.

(3) Classify the critical point at the origin of $f = x_1^2 + 2x_2^2 + 3x_3^2$, $f = x_1^2 + x_2^2 - x_3^2$ and $f = x_1 x_2$. For the last one, find the eigenvalues of the Hessian.

Solution: p. 47.

5.4 The Laplacian

Everything below works in $\mathbb{R}^d$ for any $d \geq 1$. Write $x = (x_1, \ldots, x_d)$.

First form: sum of second partials

Definition 8 (Laplacian)**.**

$$\Delta = \sum_{i=1}^{d} \frac{\partial^2}{\partial x_i^2}, \qquad \Delta f(x) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(x). \tag{21}$$

Example. *Let $f(x_1, x_2, x_3) = x_1^2 - x_2^2$. Then*

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \qquad \frac{\partial^2 f}{\partial x_2^2} = -2, \qquad \frac{\partial^2 f}{\partial x_3^2} = 0, \tag{22}$$

so $\Delta f = 2 - 2 + 0 = 0$. A function with $\Delta f = 0$ is called harmonic.

Second form: divergence of the gradient

Definition 9 (Divergence)**.** *For a vector field $\vec{v}(x) = (v^1(x), \ldots, v^d(x))$, where the superscripts label components and are not powers,*

$$\nabla \cdot \vec{v} = \operatorname{div} \vec{v} = \sum_{i=1}^{d} \frac{\partial v^i}{\partial x_i}(x). \tag{23}$$

Applying this to the gradient field $\vec{v} = \nabla f$, whose i-th component is $\partial f/\partial x_i$, gives

$$\nabla \cdot (\nabla f) = \sum_{i=1}^{d} \frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_i}\right) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2} = \Delta f. \tag{24}$$

Proposition 2.

$$\Delta = \nabla \cdot \nabla = \operatorname{div} \circ \operatorname{grad}. \tag{25}$$

Remark 6. *This second form is the useful one for radial functions: combining it with the radial chain rule turns Δ into an ODE in r.*

5.5 Harmonic Radial Functions

Proposition 3 (Radial Laplacian)**.** *For $g(x) = f(r)$, $r = \|x\|$ in $\mathbb{R}^d$, $x \neq 0$:*

$$\Delta g = f''(r) + \frac{d-1}{r} f'(r). \tag{26}$$

Solve $\Delta f = 0$ in $d = 3$ with the ansatz $f(r) = r^a$:

$$\Delta f = a(a-1)r^{a-2} + \frac{2}{r}\, a r^{a-1} = a(a+1)\, r^{a-2}, \tag{27}$$

which vanishes for $r > 0$ iff $a = 0$ or $a = -1$. So

$$g(x) = A + \frac{B}{\|x\|}. \tag{28}$$

Remark 7. *$1/\|x\|$ is the Newtonian potential: $\nabla(1/r) = -x/\|x\|^3$, the inverse-square field. Harmonic away from the origin only. Its divergence is 0 for $x \neq 0$, yet its flux through every sphere around the origin is the same nonzero number (-4π). So all the "source" sits at one point, and no function can describe it. This is the motivation for distributions (Section 8).*

6 First Order PDEs and the Solution Operator

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6.1 The Underlying ODE

Start from the ODE

$$x'(s) = v(x(s), s), \tag{29}$$

where v is a velocity field: a particle at x at time s moves with velocity $v(x, s)$. Assume $v(x, s)$ is smooth. By the existence and uniqueness theorem there is a unique trajectory through every point, forwards and backwards in time.

6.2 The Solution Operator

Definition 10 (Solution operator)**.** *Given times t_0 and t_1, the solution operator is the map*

$$S_{t_0 \to t_1}(x_0) = x(t_1), \tag{30}$$

where $x(\cdot)$ solves the ODE with initial condition $x(t_0) = x_0$. In words: where is the particle at time t_1 if it was at x_0 at time t_0?

Proposition 4 (Identities)**.**

1. *$S_{t \to t}(x) = x$, that is $S_{t \to t} = \mathrm{Id}$.*
2. *$S_{t_1 \to t_2} \circ S_{t_0 \to t_1} = S_{t_0 \to t_2}$ (two legs or one, same endpoint).*
3. *$S_{t_1 \to t_0} = (S_{t_0 \to t_1})^{-1}$: running time backwards inverts the operator.*

Example (Constant velocity)**.** *$v = c$. Trajectories are lines $x(t) = x_0 + c(t - t_0)$, so $S_{t_0 \to t_1}(x) = x + c(t_1 - t_0)$.*

Example (Linear velocity)**.** *$v(x) = -x$. Trajectories are $x(t) = x_0 e^{-(t - t_0)}$, so $S_{t_0 \to t_1}(x) = x\, e^{-(t_1 - t_0)}$. In the (x, t) plane these are exponential curves collapsing toward $x = 0$ (the constant velocity ones are parallel lines).*

6.3 Transporting Labels: Where the PDE Comes From

Stamp each particle at time 0 with a label $g(x_0)$ that never changes. Let $u(x,t)$ be the label of the particle sitting at x at time t. That particle started at $S_{t\to 0}(x)$ (run the ODE backwards), so

$$u(x,t) = g\big(S_{t\to 0}(x)\big). \tag{31}$$

Labels are carried forward by the flow; to evaluate u we track backward to the data.

Example. *Constant velocity: $u(x,t) = g(x-ct)$, the signal translated at speed c. Linear velocity $v=-x$: $u(x,t) = g(xe^t)$.*

Deriving the PDE. u is constant along every trajectory, so by the chain rule

$$0 = \frac{d}{dt}u(x(t),t) = u_t + x'(t)\,u_x = u_t + v(x,t)\,u_x. \tag{32}$$

Since a trajectory passes through every point,

$$u_t + v(x,t)\,u_x = 0 \qquad \text{(linear transport equation)}. \tag{33}$$

Remark 8. *We wrote the solution first and got the PDE by differentiating. The method of characteristics is the reverse trip: read v off the PDE, rebuild the flow, then "flow backwards, read the label". The trajectories of v are exactly the characteristics. Formula (31) only defines u where the backward trajectory hits the data line, so the data must meet each trajectory exactly once. Units: v is length per time.*

6.4 Examples of Trajectories

Example. *The curve $\vec{x}(s) = (x(s), y(s)) = (3e^{s-2}, 4e^{s-2})$ satisfies $\vec{x}'(s) = \vec{x}(s)$, so it is a trajectory of the velocity field $\vec{v}(x,y) = (x,y)$. It passes through $\vec{x}(2) = (3,4)$ and $\vec{x}(5) = (3e^3, 4e^3)$.*

Example. *The curve $\vec{x}(s) = (3s+5, 5s+6)$ has constant velocity $\vec{v}(s) = \vec{x}'(s) = (3,5)$, with $\vec{x}(0) = (5,6)$ and $\vec{x}(2) = (11,16)$.*

Example (Decoupled linear field)**.** *Take $\vec{v}(x,y) = (c_1x, c_2y)$ for constants c_1, c_2, so the ODE is*

$$\begin{cases} x'(s) = c_1 x(s) \\ y'(s) = c_2 y(s) \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0. \end{cases} \tag{34}$$

The solution is $x(s) = x_0 e^{c_1 s}$, $y(s) = y_0 e^{c_2 s}$. Check: $x'(s) = c_1 x_0 e^{c_1 s} = c_1 x(s)$, and similarly for y.

Example (Rotation field)**.** *Take $\vec{v}(x,y) = (-y, x)$. A circle of radius r, $\vec{x}(s) = (r\cos s, r\sin s)$, has $\vec{x}'(s) = (-r\sin s, r\cos s) = (-y, x)$, so circles about the origin are trajectories. The general solution of*

$$\begin{cases} x' = -y \\ y' = x \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \end{cases} \tag{35}$$

is

$$x(s) = x_0\cos s - y_0 \sin s, \qquad y(s) = x_0 \sin s + y_0 \cos s. \tag{36}$$

The circle above is the case $x_0 = r$, $y_0 = 0$.

6.5 Constant Coefficient Transport

Consider

$$au_x + bu_y = 0, \qquad u = u(x,y), \qquad a, b \in \mathbb{R}. \tag{37}$$

The left side is $\nabla u \cdot (a, b)$, the directional derivative of u in the direction (a, b). So the PDE says u does not change along lines parallel to (a, b).

Proposition 5 (Constancy along characteristics)**.** *Let $\vec{v} = (a, b)$ and let $\vec{x}(s) = (x(s), y(s))$ solve $\vec{x}'(s) = \vec{v}$, that is $x'(s) = a$ and $y'(s) = b$. Then $u(\vec{x}(s))$ is constant in s.*

Proof. By the chain rule,

$$\frac{d}{ds} u(x(s), y(s)) = u_x\, x'(s) + u_y\, y'(s) = au_x + bu_y = 0. \qquad \square \tag{38}$$

The trajectories of $\vec{v}$ are called *characteristics*. Since u is constant along each one, its value is determined by wherever the characteristic meets the data.

Definition 11 (Characteristic, informal)**.** *A curve along which the PDE reduces to an ODE. Knowing u at one point of it, the PDE determines u along the whole curve. (Special to first order: u at a point depends on the data at a single point. Not true for heat or Laplace.)*

Proposition 6 (General solution)**.** *For any differentiable f,*

$$u(x,y) = f(-bx + ay) \tag{39}$$

solves $au_x + bu_y = 0$.

Proof. $u_x = -b\, f'(-bx + ay)$ and $u_y = a\, f'(-bx + ay)$, so $au_x + bu_y = (-ab + ab)f'(-bx + ay) = 0$. $\square$

Remark 9. *The argument $-bx + ay$ is constant exactly along the lines of direction (a, b), which are the characteristics. When $b = 0$ the equation reduces to $au_x = 0$ and the formula gives $u(x,y) = f(ay)$, a function of y alone, as expected.*

6.6 Adding Data

Example. *Solve*

$$\begin{cases} 3u_x + 2u_y = 0 & \textit{(PDE)} \\ u(x,0) = x^3 & \textit{(IC)}, \end{cases} \tag{40}$$

where the data is given on the line $\Gamma = \{(x,y) \mid y = 0\}$.

Method 1 (Via the general solution)**.** *With* $a = 3$, $b = 2$, *the general solution is* $u(x,y) = f(-2x+3y)$. *On* $y = 0$ *this reads* $u(x,0) = f(-2x) = x^3$. *Setting* $z = -2x$, *so* $x = -z/2$, *gives* $f(z) = -z^3/8$. *Therefore*

$$u(x,y) = f(-2x+3y) = -\frac{(-2x+3y)^3}{8} = \frac{(2x-3y)^3}{8}. \tag{41}$$

Check: $u(x,0) = (2x)^3/8 = x^3$, *and* $3u_x + 2u_y = \frac{3}{8} \cdot 3(2x-3y)^2 \cdot 2 + \frac{2}{8} \cdot 3(2x-3y)^2 \cdot (-3) = 0$.

Method 2 (Via characteristics)**.** *(1)* u *is constant along curves with* $x'(s) = 3$, $y'(s) = 2$.

(2) These are the lines $x(s) = 3s + x_0$, $y(s) = 2s + y_0$.

(3) Given a target point (x,y), *find the point* $(x_0, 0)$ *on* Γ *whose characteristic reaches it. From* $y = 2s + 0$ *we get* $s = y/2$, *and then* $x = 3s + x_0$ *gives* $x_0 = x - 3y/2$.

(4) Since u *is constant along the characteristic,* $u(x,y) = u(x_0,0) = x_0^3$*:*

$$u(x,y) = \left(x - \frac{3y}{2}\right)^3 = \frac{(2x-3y)^3}{8}. \tag{42}$$

Both methods agree.

6.7 Which Data Curves Work?

A data curve works if it meets every characteristic exactly once (here: the x-axis, the y-axis, $y = x^2 \ldots$). It fails if it *is* a characteristic. For $3u_x + 2u_y = 0$, u is forced to be constant on the line $2x - 3y = 0$, so:

- nonconstant data on that line: **no solution**;
- constant data c on that line: **infinitely many** ($u = f(2x-3y)$ with any f having $f(0) = c$).

Either way the problem is not well-posed.

Problem 6 (A1 Q3, Choksi 2.3)**.** *Every solution of* $au_x + bu_y = 0$ *satisfies* $u(1,2) = u(3,6)$. *Find* a/b. Solution: p. 49.

Problem 7 (A1 Q3, Choksi 2.5)**.** *Solve* $4u_x - 3u_y = 0$ *with* $u = e^x$ *on the line* $y = 5x$. Solution: p. 50.

7 Variable Coefficients and Inhomogeneous Transport

September 16, 2026

7.1 Warm-up: Curved Characteristics

Example. *$u_x + y\,u_y = 0$. Now the direction of zero change $(1, y)$ varies from point to point. Follow curves tangent to it: $\frac{dy}{dx} = y$, so $y = Ce^x$. Along such a curve $\frac{d}{dx}u(x, y(x)) = u_x + yu_y = 0$, so u is constant on each exponential curve. Each curve is labelled by $C = ye^{-x}$, so*

$$u(x, y) = f(ye^{-x}). \tag{43}$$

Problem 8 (A1 Q3, Choksi 2.4)**.** *For $u_x + y^2 u_y = 0$ the following values are known:*

$$u(3, 2) = 7, \quad u(4, 2) = -6, \quad u(8, 1) = -2, \quad u(6, -1) = 3,$$
$$u(6, -2) = 0, \quad u(10, 7) = 8, \quad u(15, -4) = 10.$$

Which of $u(\frac{5}{2}, \frac{1}{2})$, $u(8, -\frac{2}{5})$, $u(\frac{9}{2}, 1)$ are determined? Solution: p. 49.

Problem 9 (A1 Q4, Choksi 2.21)**.** *A solution $u(x, y)$ is constant on every curve $y = x^2 + 2x + C$. Find a first order linear PDE it satisfies.* Solution: p. 50.

7.2 Problem Setup

Problem 10 (Linear transport with variable coefficients)**.** *Inputs:*

(1) A curve $\Gamma \subset \mathbb{R}^2$ and data $u_0 : \Gamma \to \mathbb{R}$.

(2) Coefficients $a, b : \mathbb{R}^2 \to \mathbb{R}$.

Find $u : \mathbb{R}^2 \to \mathbb{R}$ such that

$$\begin{cases} a(x, y)\,u_x + b(x, y)\,u_y = 0 & \text{(PDE)} \\ u(x, y) = u_0(x, y) \quad \forall (x, y) \in \Gamma & \text{(IC)}. \end{cases} \tag{44}$$

7.3 Method of Characteristics

Method 3 (Homogeneous case)**.** *(1) Solve the characteristic ODE*

$$\begin{cases} x'(s) = a(x(s), y(s)) \\ y'(s) = b(x(s), y(s)) \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \end{cases} \qquad (x_0, y_0) \in \Gamma. \tag{45}$$

(2) Set $z(s) = u(x(s), y(s))$. By the chain rule

$$z'(s) = u_x\,x' + u_y\,y' = au_x + bu_y = 0, \tag{46}$$

so z is constant and $z(s) = u_0(x_0, y_0)$.

(3) Given a target (x, y), find s_0 and $(x_0, y_0) \in \Gamma$ with $x(s_0) = x$, $y(s_0) = y$. Then

$$u(x, y) = u_0\big(x_0(x, y),\, y_0(x, y)\big). \tag{47}$$

7.4 Inhomogeneous Case

Now allow a right hand side:

$$a(x,y)\,u_x + b(x,y)\,u_y = c_1(x,y)\,u + c_2(x,y). \tag{48}$$

The characteristics are unchanged, but u is no longer constant along them. With $z(s) = u(x(s), y(s))$,

$$z'(s) = au_x + bu_y = c_1 z + c_2, \tag{49}$$

so the method becomes a three-dimensional ODE system (the characteristic equations):

$$\begin{cases} x' = a(x,y) \\ y' = b(x,y) \\ z' = c_1(x,y)\,z + c_2(x,y) \end{cases} \qquad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \\ z(0) = u_0(x_0, y_0). \end{cases} \tag{50}$$

Because the PDE is linear, the first two equations do not involve z: solve them first, then the z equation. If $c_1 = c_2 = 0$ we are back to the homogeneous case.

Remark 10 (Vocabulary)**.** *The curves $(x(s), y(s))$ in the plane are the* projected characteristics*. The curve $(x(s), y(s), z(s))$ lies on the solution surface $z = u(x,y)$, and its shadow in the plane is the projected characteristic.*

Method 4 (Recipe)**.** *(1) Solve for $x(s), y(s)$ starting on Γ; one free parameter (e.g. x_0) says where on Γ you started.*

(2) Solve for $z(s)$ with $z(0) =$ data at that starting point.

(3) Given (x,y), solve for s and x_0, substitute into z. The final formula contains only x and y. (This is the backward flow $S_{t\to 0}$ again.)

7.5 Worked Example

Example. *Solve*

$$\begin{cases} -y\,u_x + x\,u_y = u & \textit{(PDE)} \\ u(x,0) = g(x) & \textit{(IC)}, \quad \Gamma = \{(x,0) \mid x > 0\}. \end{cases} \tag{51}$$

Solution 1. *Here $a = -y$, $b = x$, $c_1 = 1$, $c_2 = 0$.*
Step 1: characteristic system.

$$x' = -y, \qquad y' = x, \qquad z' = z, \qquad x(0) = x_0,\ y(0) = 0,\ z(0) = g(x_0). \tag{52}$$

Step 2: solve. *From the rotation field example, the general solution of $x' = -y$, $y' = x$ is*

$$x(s) = a_1 \cos s - a_2 \sin s, \qquad y(s) = a_1 \sin s + a_2 \cos s. \tag{53}$$

The initial conditions give $a_1 = x_0$, $a_2 = 0$, so

$$x(s) = x_0 \cos s, \qquad y(s) = x_0 \sin s, \qquad z(s) = g(x_0)\, e^s. \tag{54}$$

Step 3: invert. *Require $x(s_0) = x$, $y(s_0) = y$:*

$$x = x_0 \cos s_0, \quad y = x_0 \sin s_0 \quad \Longrightarrow \quad x_0 = \sqrt{x^2 + y^2}, \qquad s_0 = \arctan(y/x). \tag{55}$$

Step 4: read off *u.*

$$u(x, y) = z(s_0) = g\Big(\sqrt{x^2 + y^2}\Big) \exp\big(\arctan(y/x)\big). \tag{56}$$

In polar coordinates this is $u = g(r)\, e^{\theta}$, and the PDE is $\partial_\theta u = u$, which checks.

Remark 11. *The characteristics are circles centred at the origin, so every point except the origin is reached from Γ. At the origin both coefficients vanish and the characteristic direction degenerates, so u is undefined there. The formula with* arctan *is valid for $x > 0$; elsewhere use the polar angle $\theta \in [0, 2\pi)$ measured from the positive x-axis. Note u jumps by a factor $e^{2\pi}$ across Γ itself when $g \neq 0$, since going once around the circle multiplies z by $e^{2\pi}$.*

Problem 11 (A1 Q4, Choksi 2.9)**.** *Solve $yu_x + 3u_y = -u$, $u(x, 0) = \sin x$.* Solution: p. 50.

Problem 12 (A1 Q4, Choksi 2.10)**.** *Show that $u_x + xu_y = 0$, $u(x, 0) = e^x$ has no solution in any neighbourhood of the origin, and sketch the characteristics to see why.* Solution: p. 50.

7.6 Time as the Parameter

If one variable is time, data is at $t = 0$, and the coefficient of u_t is 1,

$$u_t + a(x, t)\, u_x = \dots, \tag{57}$$

then the characteristic equation for t is $t' = 1$, $t(0) = 0$, so $t = s$. Drop s and use t itself as the parameter: characteristics are curves $x(t)$ in the (x, t) plane. This is exactly the setting of the solution operator section.

Example (Variable speed transport)**.**

$$u_t - x\, u_x = 0, \qquad u(x, 0) = \frac{1}{1 + x^2}. \tag{58}$$

Characteristics: $x'(t) = -x$, $x(0) = x_0$, so $x(t) = x_0 e^{-t}$, and $z' = 0$ so $z = \frac{1}{1+x_0^2}$. Invert: $x_0 = xe^t$ (this is $S_{t\to 0}$). Therefore

$$u(x, t) = \frac{1}{1 + x^2 e^{2t}}. \tag{59}$$

Every point of the signal moves toward 0 at speed $|x|$, so the blip narrows while its height stays 1. Variable speed distorts the shape; constant speed only translates it.

Problem 13 (A1 Q3, Choksi 2.6)**.** *Solve $u_t + xu_x = 0$, $u(x,0) = \frac{1}{1+x^2}$, once by the backward flow formula and once by the method of characteristics, and sketch the profiles. (The mirror image of the variable speed example: velocity $v = x$ instead of $-x$.)* Solution: p. 50.

8 Distributions A: The Delta "Function" Is Not a Function

September 16, 2026

Source: Choksi Ch. 5. Goal: a weaker notion of solving, e.g. $u'' = f$, that allows a right hand side like a point source. The one idea to keep: **pointwise values are the wrong data; integrals against test functions are the right data.**

8.1 Integrable Functions

Definition 12. *$f : \mathbb{R} \to \mathbb{R}$ is* integrable *if $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$. It is* locally integrable *if $\int_a^b |f(x)|\,dx < \infty$ for all $a < b$.*

Example.

- *$f(x) = 1$ (or any polynomial): locally integrable, $\int_a^b 1\,dx = b - a$, but not integrable.*
- *If f is continuous and $f(x) = 0$ for $|x| > R$: integrable.*
- *$\frac{1}{1+x^2}$: integrable (total integral π).*
- *$\frac{1}{\sqrt{|x|}}$: locally integrable, since $\int_{-1}^{1} |x|^{-1/2} dx = 4$ is finite.*
- *$\frac{1}{x}$: not locally integrable, $\int_0^1 \frac{dx}{x} = \log x\big|_0^1 = \infty$.*

Remark 12 (Paradigm)**.** *If $f(x) = g(x)$ for all $x \neq 3$ but $f(3) \neq g(3)$, then $\int f = \int g$ over every interval, and $\int f\phi = \int g\phi$ for every ϕ. Integration cannot see single points, so we treat f and g as the same function.*

8.2 What We Want from δ_0

Informally, δ_0 is a "point mass" at 0: $\delta_0(x) = 0$ for $x \neq 0$ but $\int_{-\infty}^{\infty} \delta_0(x)\,dx = 1$. No function $\delta_0 : \mathbb{R} \to \mathbb{R}$ can do this (by the paradigm, a function that is 0 except at one point has integral 0). What we really want is: for every continuous f,

$$\text{“}\int_{-\infty}^{\infty} \delta_0(x) f(x)\,dx\text{”} = f(0). \tag{60}$$

8.3 First Appearance: the Derivative of Heaviside

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases} \tag{61}$$

Pointwise $H'(x) = 0$ for $x \neq 0$ and undefined at 0. Ignoring the point gives $H' = 0$, but then H would be constant, which it is not. The whole jump happens at 0. **The computation that reveals H'.** Fix ϕ smooth, vanishing for large $|x|$, and set

$$F(a) = \int_{-\infty}^{\infty} H(x-a)\phi(x)\,dx = \int_{a}^{\infty} \phi(x)\,dx. \tag{62}$$

- Differentiating under the integral: $F'(a) = -\int H'(x-a)\phi(x)\,dx$. If $H' = 0$ this gives 0.
- Directly (fundamental theorem of calculus): $F'(a) = -\phi(a)$. Not zero.

So whatever H' is, $\int H'(x-a)\phi(x)\,dx = \phi(a)$: it picks out the value of ϕ where the jump sits. That is exactly δ at a. **Same idea with a ramp (in class).** Smooth out the jump:

$$g_\epsilon(x) = \begin{cases} 0, & x < 0 \\ x/\epsilon, & 0 \leq x \leq \epsilon \\ 1, & x \geq \epsilon \end{cases} \qquad g'_\epsilon(x) = \begin{cases} 0, & x < 0 \\ 1/\epsilon, & 0 < x < \epsilon \\ 0, & x > \epsilon \end{cases} \tag{63}$$

Then $\int g'_\epsilon(x) f(x)\,dx = \frac{1}{\epsilon}\int_0^\epsilon f(x)\,dx$, the average of f over $[0,\epsilon]$. If f is continuous near 0, this tends to $f(0)$ as $\epsilon \to 0$. As $g_\epsilon \to H$, the derivatives act like δ_0.

8.4 Second Appearance: Spikes

Define triangles of base $2/n$ and height n (area 1):

$$f_n(x) = \begin{cases} n - n^2|x|, & |x| \leq 1/n \\ 0, & |x| > 1/n. \end{cases} \tag{64}$$

Pointwise, $f_n(x) \to 0$ for $x \neq 0$ and $f_n(0) = n \to \infty$: useless, and many sequences with total mass other than 1 have the same pointwise limit. But for continuous ϕ, $\phi \approx \phi(0)$ on the tiny interval $[-1/n, 1/n]$, so

$$\int f_n \phi\,dx \approx \phi(0)\int f_n\,dx = \phi(0) \quad \Longrightarrow \quad \lim_{n\to\infty}\int f_n\phi\,dx = \phi(0). \tag{65}$$

Example (In class)**.** $\phi = 1$*:* $\int f_n = 1$*.* $\phi = x$*:* $\int f_n x\,dx = 0$ *(f_n even, x odd).* $\phi = x^2$*:*

$$2\int_0^{1/n} (nx^2 - n^2x^3)\,dx = 2\left(\frac{1}{3n^2} - \frac{1}{4n^2}\right) = \frac{1}{6n^2} \to 0. \tag{66}$$

Each limit equals $\phi(0)$ (1, 0, 0), as predicted. This is Dirac's picture: a huge spike of area 1, shape irrelevant.

Remark 13 (Punchline)**.** *δ appears when differentiating H and as the limit of concentrating functions. Both times pointwise values fail and integrating against a generic ϕ succeeds. So we will* define *objects by what they do to test functions.*

9 Distributions B: Distributions and Their Derivatives

September 21, 2026

Source: Choksi 5.3, 5.4.

9.1 Motivation: Weak Solutions of $u'' = f$

A classical solution of $u''(x) = f(x)$ needs $u \in C^2$, so f must be continuous. For example $u'' = -1$ on $(-1, 1)$ with $u(-1) = u(1) = 0$ has $u(x) = \frac{1-x^2}{2}$. We want to allow a point source: replace f by concentrating f_ϵ, solve

$$u_\epsilon'' = f_\epsilon, \qquad u_\epsilon(-1) = u_\epsilon(1) = 0, \tag{67}$$

and let $\epsilon \to 0$. To make sense of the limit equation $u'' = \delta_0$ we need distributions.

9.2 Test Functions

Definition 13 (Support)**.** *The support of a continuous ϕ is the closure of $\{x : \phi(x) \neq 0\}$. ϕ has* compact support *if $\phi = 0$ outside some interval $[-R, R]$.*

Definition 14 (Test functions)**.** *$C_c^\infty(\mathbb{R})$ is the set of infinitely differentiable functions with compact support.*

No polynomial works (a polynomial vanishing on an interval is 0). The standard example is the **bump function**

$$\phi(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1 \\ 0, & |x| \geq 1, \end{cases} \tag{68}$$

with support $[-1, 1]$. The only issue is smoothness at $x = \pm 1$: every derivative tends to 0 there because the exponential beats every power (Exercise 5.1).

Building more test functions from one: translates and rescalings $\phi(ax + b)$, sums and scalar multiples, products $\psi\phi$ with ψ any C^∞ function, and most importantly **derivatives**: $\phi \in C_c^\infty(\mathbb{R}) \Rightarrow \phi^{(k)} \in C_c^\infty(\mathbb{R})$ for all k. This is why we insist on C^∞.

9.3 Definition of a Distribution

Definition 15 (Distribution)**.** *A distribution F is a rule assigning to each test function ϕ a real number,*

$$F : C_c^\infty(\mathbb{R}) \to \mathbb{R}, \qquad \phi \mapsto \langle F, \phi \rangle \in \mathbb{R}, \tag{69}$$

which is

- ***linear**: $\langle F, a\phi + b\psi \rangle = a\langle F, \phi \rangle + b\langle F, \psi \rangle$,*
- ***continuous**: $\phi_n \to \phi$ in $C_c^\infty(\mathbb{R})$ implies $\langle F, \phi_n \rangle \to \langle F, \phi \rangle$.*

A map from functions to numbers is called a functional*. $\langle F, \phi \rangle$ is just notation for $F(\phi)$, chosen to look like an integral.*

Definition 16 (Convergence in $C_c^\infty(\mathbb{R})$)**.** *$\phi_n \to \phi$ in $C_c^\infty(\mathbb{R})$ if*

(i) all ϕ_n and ϕ vanish outside one common $[-R, R]$, and

(ii) for every $k \geq 0$, $\max_x |\phi_n^{(k)}(x) - \phi^{(k)}(x)| \to 0$ (every derivative converges uniformly).

Distributions form a vector space: $\langle F+G, \phi \rangle = \langle F, \phi \rangle + \langle G, \phi \rangle$ and $\langle cF, \phi \rangle = c\langle F, \phi \rangle$.

Remark 14 (Rule of the game)**.** *Never ask for the value of a distribution at a point x. Ask for its value on a test function ϕ: think of ϕ as a probe.*

9.4 Functions as Distributions

Proposition 7. *If f is locally integrable, then*

$$\langle F_f, \phi \rangle = \int_{-\infty}^{\infty} f(x)\phi(x)\, dx \tag{70}$$

defines a distribution F_f, the distribution generated by *f.*

Proof. If $\phi = 0$ outside $[-R, R]$, then $\left|\int f\phi\right| \leq \max|\phi| \int_{-R}^{R} |f| < \infty$, so it is well defined (this is exactly why we need local integrability). Linearity is linearity of the integral. Continuity: with common support $[-R, R]$,

$$|\langle F_f, \phi_n \rangle - \langle F_f, \phi \rangle| \leq \max|\phi_n - \phi| \int_{-R}^{R} |f|\, dx \to 0. \qquad \square \tag{71}$$

Example (5.3.1)**.** *$f \equiv 1$: $\langle F_1, \phi \rangle = \int \phi\, dx$. Likewise $f \equiv 3$: $\langle F_3, \phi \rangle = 3\int \phi\, dx$.*

Remark 15. *Functions that differ only on a set integration cannot see (measure zero) generate the same distribution; otherwise F_f determines f. Notation: we usually just write f for F_f. "In the sense of distributions" means the statement is about F_f.*

9.5 The Delta Function

Definition 17 (5.3.2). *$\langle \delta_0, \phi \rangle = \phi(0)$, and for $a \in \mathbb{R}$, $\langle \delta_a, \phi \rangle = \phi(a)$.*

It is linear, and continuous since $\phi_n \to \phi$ uniformly gives $\phi_n(0) \to \phi(0)$.

- There is no integral in the definition. Never write $\delta_0(x)$ or $\int \delta_0(x)\phi(x)\,dx$.
- It is not a function:

Proposition 8. *There is no locally integrable f with $F_f = \delta_0$.*

Proof. Suppose $\int f\phi = \phi(0)$ for every test function. Take $\phi = e \cdot (\text{bump})$, so $\phi(0) = 1$, $0 \le \phi \le 1$, support $[-1, 1]$. Let $\phi_n(x) = \phi(nx)$: support $[-1/n, 1/n]$, $\phi_n(0) = 1$. Then

$$1 = \phi_n(0) = \int f\phi_n\,dx \le \int_{-1/n}^{1/n} |f|\,dx \to 0, \tag{72}$$

since the integral of a locally integrable function over a shrinking interval goes to 0. Contradiction. □

Problem 14 (A2 Q5, Choksi 5.11). *Is $\sum_{n=1}^{\infty} \delta_n$ a well-defined distribution? If yes, prove it. Then give a function f whose derivative in the sense of distributions is $\sum_{n=1}^{\infty} \delta_n$.* Solution: p. 52.

9.6 Derivative of a Distribution

Motivation. For $f \in C^1$ and a test function ϕ, integrate by parts (boundary terms vanish by compact support):

$$\int f'\phi\,dx = -\int f\phi'\,dx. \tag{73}$$

The derivative moves onto ϕ at the cost of a sign, and the right side makes sense even if f is not differentiable.

Definition 18 (5.4.1).

$$\langle F', \phi \rangle = -\langle F, \phi' \rangle. \tag{74}$$

This is a distribution (ϕ' is a test function; linear; continuous since $\phi_n \to \phi$ implies $\phi_n' \to \phi'$). So **every distribution, hence every locally integrable function, has a derivative**.

Proposition 9 (Consistency, 5.20). *If $f \in C^1$, then $(F_f)' = F_{f'}$: the distributional derivative agrees with the classical one. (Proof: the integration by parts above.)*

Example. $\langle \delta_0', \phi \rangle = -\phi'(0)$. *Higher derivatives, one sign per derivative:*

$$\langle F^{(k)}, \phi \rangle = (-1)^k \langle F, \phi^{(k)} \rangle. \tag{75}$$

Remark 16. *Like the transpose in linear algebra ($\langle Ax, y \rangle = \langle x, A^T y \rangle$): the transpose of $\frac{d}{dx}$ is $-\frac{d}{dx}$. Every operation on distributions will be defined this way.*

9.7 Derivatives of Piecewise Smooth Functions

Example (Heaviside, 5.4.1)**.**

$$\langle H', \phi\rangle = -\int_{-\infty}^{\infty} H\phi'\,dx = -\int_{0}^{\infty} \phi'\,dx = -\big(0-\phi(0)\big) = \phi(0), \tag{76}$$

using $\phi(\infty)=0$. So $H'=\delta_0$ in the sense of distributions. (We never wrote $\int \delta_0(x)\phi(x)dx$.)

Remark 17 (Quiz hint, September 28)**.** *Be able to show $H'=\delta_0$ from the definition, exactly as above.*

Example ($|x|$, 5.4.2)**.** *Split at 0 and integrate by parts on each side:*

$$\langle |x|', \phi\rangle = -\int |x|\phi'\,dx = \int_{-\infty}^{0} x\phi'\,dx - \int_{0}^{\infty} x\phi'\,dx = -\int_{-\infty}^{0}\phi\,dx + \int_{0}^{\infty}\phi\,dx, \tag{77}$$

(boundary terms vanish: at 0 because of the factor x, at $\pm\infty$ by compact support). So $|x|' = \operatorname{sgn}(x)$ (-1 for $x<0$, $+1$ for $x>0$); the value at 0 is irrelevant because $|x|$ is continuous. Second derivative:

$$\langle \operatorname{sgn}', \phi\rangle = -\int \operatorname{sgn}\,\phi' = \int_{-\infty}^{0}\phi' - \int_{0}^{\infty}\phi' = \phi(0)+\phi(0) = 2\phi(0), \tag{78}$$

so $|x|'' = 2\delta_0$. The jump of size 2 in sgn *shows up as $2\delta_0$.*

Theorem 2 (Derivative of a function with a jump)**.** *Let f be C^1 except at one point a, with f and f' having one-sided limits at a. Let $\{f'\}$ denote the pointwise derivative for $x\neq a$. Then, in the sense of distributions,*

$$f' = \{f'\} + \big(f(a^+) - f(a^-)\big)\,\delta_a. \tag{79}$$

In particular, if f is continuous at a, then $f'=\{f'\}$.

Proof idea. Split $-\int f\phi'$ at a and integrate by parts on each side. The interior terms give $\int\{f'\}\phi$; the boundary terms at a give $f(a^+)\phi(a) - f(a^-)\phi(a)$. □

Checks: H has jump 1 at 0, $\{H'\}=0$, so $H'=\delta_0$. $|x|$ has no jump, so $|x|' = \operatorname{sgn}$.

September 28, 2026

Example. *Let*

$$f(x) = \begin{cases} 2x+3, & x<0 \\ x^2, & x\geq 0. \end{cases} \tag{80}$$

Remove the jump: $f = g - 3H$, where

$$g(x) = \begin{cases} 2x+3, & x<0 \\ x^2+3, & x\geq 0 \end{cases} \tag{81}$$

is continuous and piecewise differentiable, so $g' = \{g'\}$. *With* $(-3H)' = -3\delta_0$,

$$f' = \{f'\} - 3\delta_0, \qquad \{f'\}(x) = \begin{cases} 2, & x < 0 \\ 2x, & x > 0, \end{cases} \tag{82}$$

matching the theorem with jump $f(0^+) - f(0^-) = 0 - 3 = -3$.

Method 5 (Recipe). *(1) Fix a test function* ϕ *and write* $\langle f', \phi\rangle = -\int f\phi'$.

(2) Split $\mathbb{R}$ *into intervals where* f *is smooth.*

(3) Integrate by parts on each piece. Boundary terms at $\pm\infty$ *vanish; boundary terms at a singularity add up to (jump of* f*)* $\times\, \phi(a)$*, i.e. a delta.*

In class: Choksi Exercises 5.2 and 5.8.

Problem 15 (A2 Q1, Choksi 5.2). *Let* $f(x) = \max(x, 0)$. *Find* f' *and* f'' *in the sense of distributions.* Solution: p. 51.

Problem 16 (A2 Q2, Choksi 5.3). *Let* $f(x) = |x|$ *on* $(-1, 1]$, *extended to* $\mathbb{R}$ *with period* 2. *Find* f' *and* f'' *in the sense of distributions, and sketch* f, f', f''. Solution: p. 51.

Problem 17 (A2 Q3, Choksi 5.4). *Give a function* f *whose first and second derivatives in the sense of distributions are generated by functions, but whose third derivative is* δ_0. Solution: p. 52.

Problem 18 (A2 Q4, Choksi 5.9 (sample, not submitted)). *Find a function* f *whose derivative in the sense of distributions is* $x^2 + 4x + \delta_2$. Solution: p. 52.

Problem 19 (A2 Q4, several jumps (sample, not submitted)). *Let* f *be* C^1 *on* $\mathbb{R}$ *except at* $x_1 < \cdots < x_k$, *with* f *and* f' *having one-sided limits at each* x_j. *Let* $g = f'$ *away from the* x_j. *Prove that in the sense of distributions*

$$f' = g + \sum_{j=1}^{k} \big(f(x_j^+) - f(x_j^-)\big)\delta_{x_j}.$$

Solution: p. 52.

9.8 Fundamental Solution of $\frac{d^2}{dx^2}$

Divide the $|x|$ example by 2:

$$\left(\frac{|x|}{2}\right)'' = \delta_0. \tag{83}$$

A solution of $u'' = \delta_0$ is a *fundamental solution*: the response to a unit source at one point. $u = |x|/2$ is continuous, has a kink at the source, and solves $u'' = 0$ away from it. The solution is two derivatives smoother than the right side.

Example. *Back to the motivation: $u'' = \delta_0$ on $(-1, 1)$ with $u(\pm 1) = 0$ is solved by $u(x) = \frac{|x|-1}{2}$ (add the constant $-\frac{1}{2}$, which has zero second derivative, to fix the boundary values).*

Remark 18. *This is the 1D version of the point mass from week 1: in $\mathbb{R}^3$, $\Delta \frac{1}{|x|} = 0$ away from 0, and in the sense of distributions $\Delta \frac{1}{|x|} = -4\pi\delta_0$. Proved later with Laplace's equation (Choksi Ch. 8).*

Assignment Problems: Fundamental Solutions

Problem 20 (A2 Q12)**.** *Let $f_\epsilon = 1/\epsilon$ for $|x| \le \epsilon/2$ and 0 otherwise. Solve $u'' = f_\epsilon$ on $[-1, 1]$ with $u(\pm 1) = 0$; call the solution u_ϵ. Show that $u_\epsilon \to \Phi$ in the sense of distributions, where $\Phi = \frac{1}{2}|x|$ (up to a constant).* Solution: p. 55.

Problem 21 (A2 Q13)**.** *Show that $\Phi_+(x) = \max(x, 0)$ and $\Phi_-(x) = \max(-x, 0)$ also satisfy $\Phi'' = \delta_0$.* Solution: p. 55.

Problem 22 (A2 Q14)**.** *Show that the difference of any two of Φ, Φ_+, Φ_- is linear. Which of the three is even? Which vanishes for $x < 0$?* Solution: p. 56.

Problem 23 (A2 Q15)**.** *Let f be continuous with compact support and $u(x) = \int_{-\infty}^{\infty} \Phi(x - y) f(y)\, dy$. Split the integral at $y = x$, differentiate twice, and show $u'' = f$ classically (the 1D Poisson equation solved by convolution with the fundamental solution).* Solution: p. 56.

Problem 24 (A2 Q16, challenge)**.** *Show that every distribution G with $G' = 0$ is a constant function. (Hint: a test function ψ is the derivative of a test function if and only if $\int \psi = 0$.) Conclude that every solution of $T'' = \delta_0$ is $\frac{1}{2}|x|$ plus a linear function.* Solution: p. 56.

10 Distributions C: Convergence in the Sense of Distributions

September 23, 2026

Source: Choksi 5.5.

10.1 Definition

Definition 19 (5.5.1)**.** *Distributions F_n converge to a distribution F* in the sense of distributions *if*

$$\langle F_n, \phi \rangle \xrightarrow{n\to\infty} \langle F, \phi \rangle \qquad \text{for all } \phi \in C_c^\infty(\mathbb{R}). \tag{84}$$

Definition 20 (5.5.2)**.** *Locally integrable f_n converge to F in the sense of distributions if $F_{f_n} \to F$, that is,*

$$\int_{-\infty}^{\infty} f_n(x)\phi(x)\, dx \to \langle F, \phi \rangle \qquad \text{for all } \phi \in C_c^\infty(\mathbb{R}). \tag{85}$$

Remark 19. *For each fixed probe ϕ this is convergence of real numbers. Different probes may converge at different rates; nothing uniform in ϕ is required. Two cases: $f_n \to f$ means $\int f_n\phi \to \int f\phi$; $f_n \to \delta_0$ means $\int f_n\phi \to \phi(0)$.*

Proposition 10 (5.5.1, derivatives converge automatically)**.** *If $F_n \to F$ in the sense of distributions, then $F_n' \to F'$, and likewise for every higher derivative.*

Proof. $\langle F_n', \phi\rangle = -\langle F_n, \phi'\rangle \to -\langle F, \phi'\rangle = \langle F', \phi\rangle$, because ϕ' is a test function. □

Remark 20. *In ordinary calculus $f_n \to f$ uniformly says nothing about f_n'. Here the transpose trick pays off: limits of distributions can always be differentiated term by term.*

10.2 Distributional versus Pointwise Convergence

Proposition 11 (5.5.2)**.** *Let f_n be locally integrable with $f_n \to f$ pointwise, f locally integrable, and suppose there is a locally integrable g with $|f_n(x)| \le g(x)$ for all n and x. Then $f_n \to f$ in the sense of distributions.*

Proof. Dominated convergence (Choksi Theorem A.8) applied to $\int f_n\phi$ on the support of ϕ. □

Example. *$f_n(x) = \frac{1}{n}e^{-x^2/4n} \to 0$ pointwise and $|f_n| \le 1$, so $f_n \to 0$ in the sense of distributions.*

Remark 21 (Neither direction is free)**.**

- *The spikes of Section 8 tend to 0 pointwise for $x \neq 0$, yet converge to δ_0, not 0. The proposition fails: any dominating g must be of size $\frac{1}{4|x|}$ near 0, which is not locally integrable.*
- $\sin(nx) \to 0$ *in the sense of distributions (Theorem below), yet* $\sin(nx)$ *does not converge for fixed x not a multiple of π.*

10.3 Approximations to the Identity

With $\sigma_n = 1/n$, four sequences:

$$\text{Hats:} \qquad f_n(x) = \tfrac{n}{2} \text{ for } |x| \le \sigma_n,\ 0 \text{ otherwise,} \tag{86}$$

$$\text{Spikes:} \qquad f_n(x) = n - n^2|x| \text{ for } |x| < \sigma_n,\ 0 \text{ otherwise,} \tag{87}$$

$$\text{Arctangent derivative:} \qquad f_n(x) = \frac{1}{\pi}\frac{\sigma_n}{x^2+\sigma_n^2}, \tag{88}$$

$$\text{Gaussians:} \qquad f_n(x) = \frac{1}{\sqrt{2\pi\sigma_n}}\,e^{-x^2/2\sigma_n}. \tag{89}$$

All four converge to δ_0 in the sense of distributions, i.e. $\int f_n\phi \to \phi(0)$. Such a sequence is an *approximation to the identity.*

Definition 21 (Properties 5.5.1)**.** *(1) Nonnegativity:* $f_n \geq 0$.

(2) Unit mass: $\int_{-\infty}^{\infty} f_n \, dx = 1$ *for every* n.

(3) Concentration at 0*: as* $n \to \infty$ *the mass of* f_n *concentrates near* $x = 0$.

Example. *Unit mass for the arctangent sequence:* $\int \frac{\sigma}{x^2+\sigma^2} \, dx = \left[\arctan(x/\sigma)\right]_{-\infty}^{\infty} = \pi$.

Remark 22 (Heuristic)**.** *By (1) and (3), almost all the mass sits in* $[-\delta, \delta]$*, where* $\phi \approx \phi(0)$*, so* $\int f_n \phi \approx \phi(0) \int_{-\delta}^{\delta} f_n \approx \phi(0)$*. Making (3) precise is what the proofs do.*

Theorem 3 (5.1)**.** *The spikes converge to* δ_0 *in the sense of distributions.*

Proof. Fix ϕ and $\epsilon > 0$. Since $\int f_n = 1$, $\phi(0) = \int f_n(x)\phi(0) \, dx$, so using $f_n \geq 0$,

$$\left| \int f_n \phi \, dx - \phi(0) \right| \leq \int f_n(x) \, |\phi(x) - \phi(0)| \, dx. \tag{90}$$

By continuity there is $\delta > 0$ with $|\phi(x) - \phi(0)| < \epsilon$ for $|x| < \delta$. Choose N with $1/N < \delta$. For $n > N$, f_n vanishes outside $[-1/n, 1/n]$, so the right side is $< \epsilon \int f_n = \epsilon$. □

Remark 23. *The proof used only (1), (2) and the support* $[-1/n, 1/n]$*, never the formula. The shape is irrelevant, as Dirac said.*

Theorem 4 (5.2, arctangent sequence)**.** *The arctangent derivatives converge to* δ_0.

Proof sketch. They are positive everywhere, so the tails must be handled. Split the integral into $|x| < \delta$ (below $\epsilon/2$ by continuity) and $|x| \geq \delta$, where $|\phi(x) - \phi(0)| \leq 2C$ with $C = \max |\phi|$ and

$$\int_{|x| \geq \delta} f_n \, dx = \frac{2}{\pi} \left(\frac{\pi}{2} - \arctan \frac{\delta}{\sigma_n} \right) \to 0. \tag{91}$$

Choose N so this is below $\epsilon/(4C)$. Full proof: Choksi pp. 178 to 180. □

Theorem 5 (5.3, shape does not matter)**.** *Let* $f \geq 0$ *be integrable with* $\int f = 1$*, and put* $f_n(x) = nf(nx)$*. Then* $f_n \to \delta_0$ *in the sense of distributions.*

Proof. Substitute $y = nx$: $\int nf(nx)\phi(x) \, dx = \int f(y)\phi(y/n) \, dy$. As $n \to \infty$, $\phi(y/n) \to \phi(0)$, and $|f(y)\phi(y/n)| \leq \max |\phi| \, |f(y)|$ is integrable, so by dominated convergence the integral tends to $\phi(0) \int f = \phi(0)$. □

Remark 24. *Nonnegativity was not used. Hats and spikes are* $nf(nx)$ *with* f *the first hat or spike; the arctangent sequence uses* $f = \frac{1}{\pi(1+x^2)}$*. The Gaussians are a squeeze by* $\sqrt{n}$ *instead of* n *(Exercise 5.13).*

Problem 25 (A2 Q6, Choksi 5.6)**.** *Prove that the hats, $f_n = \frac{n}{2}$ on $|x| \le \frac{1}{n}$ and 0 elsewhere, converge to δ_0 in the sense of distributions.* Solution: p. 53.

Problem 26 (A2 Q8, Choksi 5.14)**.** *Prove that the Gaussians converge to δ_0 in the sense of distributions, either by an ϵ versus N argument or by proving the continuum version of Theorem 5.3 (Exercise 5.13) and applying it.* Solution: p. 53.

10.4 Oscillation: $\sin(nx) \to 0$

Remark 25 (A tempting argument, and why to distrust it)**.** *For large n, ϕ is nearly constant on each period of $\sin(nx)$, so each period contributes nearly 0. But there are about n periods per unit length, so "many small pieces" needs an actual estimate (the same argument would prove every Riemann integral is 0).*

Theorem 6 (5.4)**.** $\sin(nx) \to 0$ *in the sense of distributions. The same holds for $\cos(nx)$.*

Proof. Let $\phi = 0$ for $|x| \ge L$ and $C = \max|\phi'|$. Integrate by parts using $\phi(\pm L) = 0$:

$$\left|\int_{-L}^{L} \sin(nx)\phi(x)\,dx\right| = \left|\frac{1}{n}\int_{-L}^{L} \cos(nx)\phi'(x)\,dx\right| \le \frac{2LC}{n} \to 0. \qquad \square \quad (92)$$

Remark 26. *Only compact support and one derivative of ϕ were used. Under much weaker hypotheses this is the* Riemann–Lebesgue lemma*: Fourier coefficients of integrable functions tend to 0 (Section 13).*

Example. *By Proposition 5.5.1, $(\sin nx)' = n\cos(nx) \to 0$ in the sense of distributions, although the amplitude n is unbounded. Oscillation cancels growth.*

Remark 27 (Averaging, Exercise 5.17)**.** *If f is ℓ-periodic with integral A over a period, then $f(nx) \to A/\ell$, its average, in the sense of distributions.*

Problem 27 (A2 Q7, Choksi 5.12)**.** *(a) Let $f = -1$ on $[0, \frac{1}{2}]$ and 1 on $(\frac{1}{2}, 1)$, extended with period 1, and $f_n(x) = f(nx)$. Sketch f_1, f_2, f_4. Does f_n converge in the sense of distributions, and to what? (b) Repeat with the values -1 and 5.* Solution: p. 53.

Problem 28 (A2 Q9, Choksi 5.17 (not submitted))**.** *Let f be integrable and ℓ-periodic with integral A over each period. Prove that $f_n(x) = f(nx) \to A/\ell$ in the sense of distributions.* Solution: p. 54.

10.5 Limits That Produce δ_0'

For $h > 0$,

$$\left\langle \frac{\delta_0 - \delta_h}{h}, \phi \right\rangle = \frac{\phi(0) - \phi(h)}{h} \xrightarrow{h\to 0} -\phi'(0) = \langle \delta_0', \phi \rangle. \quad (93)$$

Two opposite point sources of size $1/h$, close together, converge to a *dipole*. Equivalently, $(\delta_0 - \delta_h)/h$ is the derivative of the function equal to $1/h$ on $[0, h]$, which tends to δ_0, so Proposition 5.5.1 applies.

Problem 29 (A2 Q10, Choksi 5.18)**.** *Find the distributional limit of* $F_n = n\delta_{-1/n} - n\delta_{1/n}$. Solution: p. 54.

Problem 30 (A2 Q11, Choksi 5.19 (not submitted))**.** *Find a sequence of functions* f_n *that converges to* δ_0' *in the sense of distributions, and sketch it.* Solution: p. 54.

10.6 Preview: Sinc and Dirichlet Kernels

Two sequences that take negative values, so Properties 5.5.1 fail:

$$S_n(x) = \frac{\sin(nx)}{\pi x}, \qquad K_n(x) = \frac{\sin\left((n+\frac{1}{2})x\right)}{\sin(x/2)}, \tag{94}$$

with $S_n(0) = n/\pi$, $K_n(0) = 2n+1$. Both have $\int_{\mathbb{R}} S_n = 1$ and $\int_{-\pi}^{\pi} K_n = 2\pi$, and

$$S_n \to \delta_0 \text{ on } \mathbb{R}, \qquad K_n \to 2\pi\delta_0 \text{ on } (-\pi, \pi). \tag{95}$$

The tails of K_n oscillate with amplitude that does not tend to 0, so continuity of ϕ is not enough. This is why pointwise convergence of Fourier series is delicate (Section 13).

10.7 Summary

- A distribution is defined by its action on test functions. Never ask for its value at a point.
- Every distribution has a derivative, $\langle F', \phi\rangle = -\langle F, \phi'\rangle$. A jump gives a delta with the jump as coefficient. To remember: $H' = \delta_0$ and $(|x|/2)'' = \delta_0$.
- Convergence is tested one probe at a time. It captures concentration (limit δ_0) and oscillation (limit the average), and derivatives of convergent sequences converge.
- Six sequences converge to δ_0: hats, spikes, arctangent derivatives, Gaussians, sinc functions, Dirichlet kernels. The Gaussians solve the diffusion equation; the last two run Fourier analysis.

11 Fourier Series A: Orthogonality Gives the Formulas

September 30, 2026

Source: Choksi 11.1 to 11.3. Notation: in Chapter 5, ϕ was a test function. Here, following Choksi, ϕ is the function being expanded.

Fourier's claim (1807, memoir on heat flow): almost any function on an interval is a sum of sines and cosines. Lagrange objected that smooth sines cannot add up to a corner or jump. Yet the partial sums

$$S_N(x) = \frac{4}{\pi} \sum_{\substack{n \le N \\ n \text{ odd}}} \frac{1}{n} \sin\left(\frac{n\pi x}{l}\right) \tag{96}$$

flatten toward 1 on $(0, l)$, toward -1 on $(-l, 0)$, and pass through 0 at $x = 0$: smooth waves building a jump. Choksi's three questions:

(1) Does it work? Do the partial sums converge to the function? (Section 13)

(2) What are the coefficients? (this section)

(3) What is special about sines and cosines? (Section 12)

The one idea: **in an orthogonal basis, each coefficient is a single inner product, and no linear system has to be solved.**

11.1 Coefficients in an Orthogonal Basis of $\mathbb{R}^N$

Definition 22 (Orthogonal basis)**.** *$\vec{x}, \vec{y} \in \mathbb{R}^N$ are orthogonal if $\vec{x} \cdot \vec{y} = 0$. A basis $\{\vec{v}_1, \ldots, \vec{v}_N\}$ is orthogonal if $\vec{v}_i \cdot \vec{v}_j = 0$ for all $i \neq j$.*

Write $\vec{x} = \sum_i c_i \vec{v}_i$ and dot with $\vec{v}_j$; every term with $i \neq j$ vanishes:

$$\vec{x} \cdot \vec{v}_j = c_j \|\vec{v}_j\|^2, \qquad c_j = \frac{\vec{x} \cdot \vec{v}_j}{\|\vec{v}_j\|^2}. \tag{97}$$

Each $c_j \vec{v}_j = \text{Proj}_{\vec{v}_j} \vec{x}$, so every vector is the sum of its projections onto an orthogonal basis.

Remark 28. *In a non-orthogonal basis, dotting gives an $N \times N$ linear system with matrix entries $\vec{v}_i \cdot \vec{v}_j$; the unknowns are coupled. Orthogonality makes the matrix diagonal. With infinitely many unknowns, a diagonal system is the only kind we can solve by hand.*

Example (In class)**.** *$\vec{v}_1 = (1, 1, 1)$, $\vec{v}_2 = (1, -1, 0)$, $\vec{v}_3 = (1, 1, -2)$ are mutually orthogonal. For $\vec{x} = (2, -3, 5)$, (97) gives*

$$c_1 = \frac{4}{3}, \qquad c_2 = \frac{5}{2}, \qquad c_3 = \frac{-11}{6}, \tag{98}$$

and indeed $\frac{4}{3}(1, 1, 1) + \frac{5}{2}(1, -1, 0) - \frac{11}{6}(1, 1, -2) = (2, -3, 5)$.

11.2 Functions as Vectors

Think of f on (a, b) as a vector with one component $f(x)$ per point. Sampling at N points with spacing Δx, the dot product times Δx is a Riemann sum:

$$\sum_{i=1}^{N} f(x_i) g(x_i)\, \Delta x \xrightarrow{N \to \infty} \int_a^b f(x) g(x)\, dx. \tag{99}$$

Definition 23 (Inner product, norm, orthogonality)**.** *For real functions on* (a, b)*,*

$$(f, g) = \int_a^b f(x)g(x)\,dx, \qquad \|f\| = (f, f)^{1/2}, \tag{100}$$

and f, g *are orthogonal on* (a, b) *if* $(f, g) = 0$*. The interval is part of the definition.*

Remark 29. *Unlike* $\mathbb{R}^N$*,* $(f, f) = 0$ *only forces* $f = 0$ *except on a set integration cannot see. This is the paradigm of Chapter 5 again.*

Definition 24 (Projection)**.** $\operatorname{Proj}_g f = \dfrac{(f, g)}{\|g\|^2}\, g.$

11.3 Orthogonality of the Sines

Proposition 12 (Orthogonality relations)**.** *For positive integers* $n \neq m$*,*

$$\int_0^l \sin\left(\frac{n\pi x}{l}\right) \sin\left(\frac{m\pi x}{l}\right) dx = 0, \qquad \int_0^l \sin^2\left(\frac{n\pi x}{l}\right) dx = \frac{l}{2}. \tag{101}$$

So $X_n(x) = \sin(n\pi x/l)$ *are mutually orthogonal on* $(0, l)$ *with* $\|X_n\|^2 = l/2$*.*

Lemma 7 (Exponentials integrate to zero over a period)**.** *For an integer* k*,*

$$\int_{-\pi}^{\pi} e^{ik\theta}\, d\theta = \begin{cases} 2\pi, & k = 0 \\ 0, & k \neq 0. \end{cases} \tag{102}$$

Proof. For $k \neq 0$ an antiderivative is $e^{ik\theta}/(ik)$, which is 2π-periodic, so it takes the same value at $\pm\pi$. □

Proof of the proposition. By Euler, $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$, so

$$\sin(n\theta)\sin(m\theta) = -\frac{1}{4}\left[e^{i(n+m)\theta} - e^{i(n-m)\theta} - e^{-i(n-m)\theta} + e^{-i(n+m)\theta}\right]. \tag{103}$$

Integrate over $(-\pi, \pi)$ with the lemma. Since $n + m \neq 0$, the outer terms give 0. If $n \neq m$ the middle terms give 0; if $n = m$ they give 2π each, so the total is π. The integrand is even, so over $(0, \pi)$ we get 0 or $\pi/2$. Substituting $\theta = \pi x/l$ gives 0 or $l/2$. □

Remark 30. *Euler's formula replaces the table of product-to-sum identities: any product of sines and cosines is a sum of exponentials, and integrating over a period keeps only frequency* 0*. For complex functions* $(f, g) = \int f\bar{g}$*, and the lemma says* $e^{in\theta}$*,* $n \in \mathbb{Z}$*, are orthogonal on* $(-\pi, \pi)$*. The computation explains nothing; Section 12 gives the reason.*

11.4 The Coefficient Formula

Proposition 13 (The coefficients are forced)**.** *Suppose* $\phi(x) = \sum_{m=1}^{\infty} b_m \sin(m\pi x/l)$ *on* $(0, l)$ *and the series can be integrated term by term after multiplying by* $\sin(n\pi x/l)$*. Then*

$$b_n = \frac{2}{l} \int_0^l \phi(x) \sin\left(\frac{n\pi x}{l}\right) dx, \qquad n = 1, 2, \ldots \tag{104}$$

Proof. Multiply by $\sin(n\pi x/l)$, integrate over $(0, l)$, exchange sum and integral, and use orthogonality: only $m = n$ survives, giving $\int_0^l \phi \sin(n\pi x/l)\, dx = \frac{l}{2} b_n$. □

Remark 31. *The exchange of sum and integral is an exchange of two limits; uniform convergence on* $[0, l]$ *allows it, and without some such hypothesis it can fail. The proposition runs one way:* if ϕ *has an expansion,* then *the coefficients are* (104)*. It does not say the series converges, or converges to* ϕ*.*

Definition 25 (11.1.1, Fourier sine series)**.** *For* ϕ *integrable on* $(0, l)$ *with* b_n *from* (104)*,* $\sum_{n=1}^{\infty} b_n \sin(n\pi x/l)$ *is the Fourier sine series of* ϕ *on* $(0, l)$*.*

As in $\mathbb{R}^N$: $b_n = (\phi, X_n)/\|X_n\|^2$ and $b_n X_n = \mathrm{Proj}_{X_n} \phi$. The factor $\frac{2}{l}$ is $1/\|X_n\|^2$ and nothing more.

11.5 Cosine and Full Fourier Series

On $(0, l)$, for integers $n, m \geq 0$,

$$\int_0^l \cos\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right) dx = \begin{cases} 0, & n \neq m \\ l/2, & n = m \geq 1 \\ l, & n = m = 0. \end{cases} \tag{105}$$

For $\phi(x) = \frac{1}{2} a_0 + \sum_{n \geq 1} a_n \cos(n\pi x/l)$ this forces

$$a_n = \frac{2}{l} \int_0^l \phi(x) \cos\left(\frac{n\pi x}{l}\right) dx, \qquad n = 0, 1, 2, \ldots \tag{106}$$

The $\frac{1}{2}$ in front of a_0 lets one formula cover $n = 0$, since the constant 1 has twice the squared norm of the other cosines. This is the *Fourier cosine series* (Definition 11.1.2).

On $(-l, l)$, any two different functions of $\{1, \cos(n\pi x/l), \sin(n\pi x/l)\}$ are orthogonal (a sine times a cosine is odd), with $\|1\|^2 = 2l$ and $\|\cos\|^2 = \|\sin\|^2 = l$. So the *full Fourier series* (Definition 11.1.3) is

$$\frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{l}\right) + b_n \sin\left(\frac{n\pi x}{l}\right) \right], \tag{107}$$

$$a_n = \frac{1}{l} \int_{-l}^{l} \phi \cos \frac{n\pi x}{l}\, dx \ (n \geq 0), \qquad b_n = \frac{1}{l} \int_{-l}^{l} \phi \sin \frac{n\pi x}{l}\, dx \ (n \geq 1). \tag{108}$$

Remark 32 (The interval matters)**.** *On $(0,l)$ sines and cosines are not orthogonal to each other: $\int_0^l \sin(\pi x/l)\cdot 1\,dx = 2l/\pi \neq 0$. That is why the full series lives on $(-l,l)$ and carries $\frac{1}{l}$ instead of $\frac{2}{l}$.*

Every coefficient is $(\phi, X_n)/\|X_n\|^2$, so the table below can be rebuilt from the $\|X_n\|^2$ column. The cosine and full series start with $\frac{1}{2}a_0$.

Series	Interval	X_n	$\|X_n\|^2$	Coefficients
Sine	$(0,l)$	$\sin\frac{n\pi x}{l}$, $n\geq 1$	$\frac{l}{2}$	$b_n = \frac{2}{l}\int_0^l \phi \sin\frac{n\pi x}{l}\,dx$
Cosine	$(0,l)$	1; $\cos\frac{n\pi x}{l}$	l; $\frac{l}{2}$	$a_n = \frac{2}{l}\int_0^l \phi\cos\frac{n\pi x}{l}\,dx$
Full	$(-l,l)$	1; $\cos\frac{n\pi x}{l}$, $\sin\frac{n\pi x}{l}$	$2l$; l	$a_n, b_n = \frac{1}{l}\int_{-l}^l \phi\,(\cos,\sin)\frac{n\pi x}{l}\,dx$

11.6 Examples

Example ($\phi\equiv 1$, Choksi 11.1.1)**.** *Sine series on $(0,l)$:*

$$b_n = \frac{2}{l}\int_0^l \sin\frac{n\pi x}{l}\,dx = \frac{2}{n\pi}\big(1-(-1)^n\big) = \begin{cases}\frac{4}{n\pi}, & n \text{ odd}\\ 0, & n \text{ even},\end{cases} \tag{109}$$

so the sine series of 1 is

$$\frac{4}{\pi}\sum_{m=0}^{\infty}\frac{1}{2m+1}\sin\left(\frac{(2m+1)\pi x}{l}\right), \tag{110}$$

the series at the start of this section. Cosine series: $a_0 = 2$, $a_n = 0$ for $n\geq 1$, so the series is 1. Nothing to compute, since 1 is already the $n=0$ cosine.

Remark 33. *If ϕ is a finite combination of the X_n, its Fourier series is that combination (a finite sum can always be integrated term by term). E.g. the full series of $\cos(26\pi x)-4-3\sin(\pi x)$ on $(-1,1)$ is itself: $a_{26}=1$, $a_0=-8$, $b_1=-3$ (Choksi Exercise 11.5(a)).*

Example (In class: $\phi(x)=x$ on $(0,l)$)**.** *Sine series (one integration by parts): $b_n = \dfrac{2l(-1)^{n+1}}{n\pi}$, size $1/n$. Cosine series (two integrations by parts): $a_0 = l$ and $a_n = \dfrac{2l\big((-1)^n-1\big)}{n^2\pi^2}$, i.e. $-\frac{4l}{n^2\pi^2}$ for n odd and 0 for n even, size $1/n^2$. The cosine coefficients decay faster.*

11.7 What the Series Converges To on $\mathbb{R}$

The series is defined for every $x\in\mathbb{R}$, and the terms decide its shape: each $\sin(n\pi x/l)$ is odd and $2l$-periodic, each $\cos(n\pi x/l)$ even and $2l$-periodic. So a convergent sine series is odd and $2l$-periodic whatever ϕ was, and similarly for the others.

Definition 26 (11.1.4, piecewise continuous)**.** *ϕ on (a,b) is piecewise continuous if it is continuous except at finitely many points, where the one-sided limits $\phi(x\pm) = \lim_{h\to 0^+} \phi(x \pm h)$ exist and are finite, and $\phi(a+)$, $\phi(b-)$ exist and are finite.*

Definition 27 (Extensions)**.** *For ϕ on $(0,l)$: the* odd extension *to $(-l,l)$ is $-\phi(-x)$ for $x < 0$; the* even extension *is $\phi(-x)$ for $x < 0$. The* periodic extension *of a function on $(-l,l)$ repeats it with period $2l$. The odd (even) periodic extension is the periodic extension of the odd (even) extension. Values at the joining points are left unspecified.*

Theorem 8 (Pointwise convergence of the three series)**.** *Suppose ϕ and ϕ' are piecewise continuous on the sampled interval. Let ϕ_{ext} be the odd periodic extension (sine series on $(0,l)$), the even periodic extension (cosine series on $(0,l)$), or the periodic extension (full series on $(-l,l)$). Then for every $x \in \mathbb{R}$ the series converges to*

$$\frac{1}{2}\big[\phi_{ext}(x+) + \phi_{ext}(x-)\big], \tag{111}$$

and to $\phi_{ext}(x)$ wherever ϕ_{ext} is continuous.

Taken on trust here; proved for the full series in Section 13.

Example (Back to $\phi \equiv 1$)**.** *The odd periodic extension of 1 is the square wave (1 on $(0,l)$, -1 on $(-l,0)$, period $2l$). It jumps at every multiple of l, where the average is 0. So (110) converges to the square wave, with value 0 at $x = kl$. The jumps come from the extension, not from ϕ. The cosine series uses the even extension, which is 1 everywhere.*

Remark 34. *Endpoint values are invisible: the sine series gives 0 at $x = 0, l$ whatever value we assign ϕ there, since coefficients are integrals.*

Example (A free series)**.** *At $x = l/2$, ϕ_{ext} is continuous and $\sin\big((2m{+}1)\pi/2\big) = (-1)^m$, so (110) gives*

$$1 = \frac{4}{\pi}\sum_{m=0}^{\infty}\frac{(-1)^m}{2m+1}, \qquad \textit{that is,} \qquad \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \tag{112}$$

Example ($\phi(x) = e^x$, Choksi Figure 11.5)**.** *Sine and cosine series on $(0,1)$, full series on $(-1,1)$, limits on $(-3,3)$ with no coefficient computed:*

- *Sine: the odd periodic extension jumps at $x = 0$ (from -1 to 1) and at $x = \pm 1$ (from e to $-e$). The series gives 0 there.*
- *Cosine: the even periodic extension is continuous. The series gives 1 at even integers and e at odd integers.*
- *Full: the periodic extension jumps at odd integers from e down to e^{-1}. The series gives $\frac{1}{2}(e + e^{-1}) = \cosh 1$ there.*

Remark 35 (Decay and smoothness)**.** *For $\phi = x$, the odd periodic extension is a sawtooth (jumps) and the coefficients are $O(1/n)$; the even periodic extension is a triangle wave (corners) and the coefficients are $O(1/n^2)$. Smoothness of the extension controls the decay (Section 13).*

Example (In class, Choksi 11.3(a))**.** *$\phi = 1, 2, 3, 4$ on $(0,1], (1,2], (2,3], (3,4]$, extended with period 4, full series with $l = 2$. At $x = 0$ the extension jumps from 4 to 1, so the series gives $\frac{5}{2}$. At $x = 1$: $\frac{3}{2}$. At $x = 4$ and $x = 40$ (multiples of 4): $\frac{5}{2}$. At $x = 7.4 \equiv 3.4$: 4.*

11.8 Orthogonal Is Not Yet a Basis

Proposition 14. *Nonzero, mutually orthogonal $\vec{v}_1, \ldots, \vec{v}_N \in \mathbb{R}^N$ form a basis.*

Proof. If $\sum c_i \vec{v}_i = \vec{0}$, dot with $\vec{v}_j$: $c_j \|\vec{v}_j\|^2 = 0$, so $c_j = 0$. Independent, and N independent vectors span $\mathbb{R}^N$. □

The last step counts dimensions; with functions there is nothing to count.

Example. *Remove the first sine: $\{\sin(n\pi x/l) : n \geq 2\}$ is still orthogonal, but expanding $\phi = \sin(\pi x/l)$ gives every coefficient 0. Nothing in the computation warned us.*

Remark 36. *So the formulas come from orthogonality, a calculus fact.* Completeness *(every reasonable ϕ is the sum of its projections) is a theorem, "the miracle", proved in Section 13. The orthogonality itself comes from $X_n = \sin(n\pi x/l)$ satisfying $-X_n'' = \frac{n^2\pi^2}{l^2} X_n$, $X_n(0) = 0 = X_n(l)$: the sines are eigenfunctions of a symmetric operator (Section 12).*

12 Fourier Series B: Eigenfunctions of $-\frac{d^2}{dx^2}$

October 7, 2026

Source: Choksi 11.2, 11.3; Strauss 5.3. The one idea: **orthogonality comes from symmetry, and the boundary condition is part of what makes the operator symmetric.**

12.1 Template: Symmetric Matrices

Definition 28 (Symmetric matrix, eigenvector)**.** *An $N \times N$ matrix A is symmetric if $A\vec{x} \cdot \vec{y} = \vec{x} \cdot A\vec{y}$ for all $\vec{x}, \vec{y}$ (equivalently $A = A^T$). A nonzero $\vec{v}$ is an eigenvector with eigenvalue λ if $A\vec{v} = \lambda\vec{v}$.*

Theorem 9 (11.1, Spectral Theorem)**.** *A symmetric matrix has real eigenvalues and an orthogonal basis of eigenvectors.*

Proposition 15. *If A is symmetric, $A\vec{v}_1 = \lambda_1\vec{v}_1$, $A\vec{v}_2 = \lambda_2\vec{v}_2$, $\lambda_1 \neq \lambda_2$, then $\vec{v}_1 \cdot \vec{v}_2 = 0$.*

Proof. $\lambda_1\, \vec{v}_1 \cdot \vec{v}_2 = A\vec{v}_1 \cdot \vec{v}_2 = \vec{v}_1 \cdot A\vec{v}_2 = \lambda_2\, \vec{v}_1 \cdot \vec{v}_2$, so $(\lambda_1 - \lambda_2)\, \vec{v}_1 \cdot \vec{v}_2 = 0$. □

Remark 37. *Orthogonality is easy and uses only symmetry. Having* enough *eigenvectors for a basis is the hard part. For functions: orthogonality here, completeness in Section 13.*

12.2 The Operator $\mathcal{A} = -\frac{d^2}{dx^2}$

Fix an interval $[a, b]$ and a boundary condition, and set $\mathcal{A}X = -X''$ on functions satisfying it. $\mathcal{A}$ is linear and plays the role of the matrix.

Definition 29 (Eigenfunction)**.** *λ is an eigenvalue and $X \not\equiv 0$ an eigenfunction if X satisfies the boundary condition and $\mathcal{A}X = \lambda X$, that is,*

$$X'' + \lambda X = 0. \tag{113}$$

Without a boundary condition every λ works ($\cos(\sqrt{\lambda}x)$, 1, $e^{\sqrt{-\lambda}x}$). **The boundary condition selects the eigenvalues.** Standard conditions on $[a, b]$:

- Dirichlet: $X(a) = 0 = X(b)$.
- Neumann: $X'(a) = 0 = X'(b)$.
- Periodic: $X(a) = X(b)$ and $X'(a) = X'(b)$.

The minus sign in $\mathcal{A}$ makes the eigenvalues nonnegative (below); mathematicians prefer $-\Delta$ to Δ for the same reason.

12.3 Green's Second Identity and Symmetric Boundary Conditions

By the product rule, $-X_1''X_2 + X_1X_2'' = \frac{d}{dx}\left(-X_1'X_2 + X_1X_2'\right)$. Integrating,

$$\int_a^b \left(-X_1''X_2 + X_1X_2''\right) dx = \Big[X_1X_2' - X_1'X_2\Big]_{x=a}^{x=b}. \tag{114}$$

This is *Green's second identity* in 1D, valid for all C^2 functions. The left side is $(\mathcal{A}X_1, X_2) - (X_1, \mathcal{A}X_2)$.

Definition 30 (11.2.1, symmetric boundary condition)**.** *A boundary condition is symmetric if, whenever X_1, X_2 satisfy it,*

$$X_1(b)X_2'(b) - X_1'(b)X_2(b) = X_1(a)X_2'(a) - X_1'(a)X_2(a). \tag{115}$$

Then $(\mathcal{A}X_1, X_2) = (X_1, \mathcal{A}X_2)$, and $\mathcal{A}$ is a symmetric operator.

Example. *Dirichlet is symmetric: every boundary term contains X_1 or X_2 at a or b, which is 0.*

Example (In class, Choksi 11.2)**.** *Neumann: every boundary term contains X_1' or X_2' at an end, which is 0. Periodic: each of X_1, X_1', X_2, X_2' takes the same value at a and b, so the two sides are equal. Both are symmetric.*

Example (Not symmetric, Strauss 5.3)**.** $X(a) = X(b)$, $X'(a) = 2X'(b)$. *With* $p_j = X_j(a) = X_j(b)$, *the left side is* $p_1X_2'(b) - X_1'(b)p_2$ *and the right side is twice that, so the condition fails for most pairs.*

Remark 38. *Symmetry is not a property of* $\frac{d^2}{dx^2}$ *alone. The boundary condition is part of the operator, as the entries are part of a matrix.*

12.4 Consequences of Symmetry

Theorem 10 (Orthogonality of eigenfunctions)**.** *For a symmetric boundary condition, eigenfunctions* X_1, X_2 *with eigenvalues* $\lambda_1 \neq \lambda_2$ *satisfy* $(X_1, X_2) = 0$.

Proof. $\lambda_1(X_1, X_2) = (\mathcal{A}X_1, X_2) = (X_1, \mathcal{A}X_2) = \lambda_2(X_1, X_2)$. □

This is the reason behind the orthogonality relations of Section 11, with no trigonometry and no eigenfunction found yet.

Proposition 16 (11.3.1, eigenvalues are real)**.** *For a symmetric boundary condition with real coefficients (such as the three above), every eigenvalue is real.*

Proof. Let X be a complex eigenfunction with eigenvalue λ. Conjugating, $\overline{X}'' + \bar{\lambda}\overline{X} = 0$ and $\overline{X}$ satisfies the same condition. Apply (114) with $X_1 = X$, $X_2 = \overline{X}$: the right side is 0 by symmetry, and the left side is

$$\int_a^b \left(\lambda X\overline{X} - \bar{\lambda}X\overline{X}\right) dx = (\lambda - \bar{\lambda}) \int_a^b |X|^2 \, dx. \tag{116}$$

Since $X \not\equiv 0$, $\lambda = \bar{\lambda}$. □

With λ real, the real and imaginary parts of X are real eigenfunctions, so from now on eigenfunctions are real.

Proposition 17 (No negative eigenvalues, Green's first identity)**.** *For Dirichlet, Neumann, or periodic conditions, every eigenvalue satisfies* $\lambda \geq 0$, *with* $\lambda = 0$ *only if* X *is constant.*

Proof. Multiply $-X'' = \lambda X$ by X and integrate by parts:

$$\lambda \int_a^b X^2 \, dx = -\int_a^b X''X \, dx = \int_a^b (X')^2 \, dx - \Big[X'X\Big]_{x=a}^{x=b}. \tag{117}$$

The boundary term vanishes (a factor is 0 at each end for Dirichlet and Neumann; equal values cancel for periodic), so $\lambda = \int (X')^2 / \int X^2 \geq 0$. □

12.5 Dirichlet Eigenfunctions

Take $X'' + \lambda X = 0$, $X(0) = 0 = X(l)$. We know $\lambda \geq 0$.

- $\lambda = 0$: X is constant, and $X(0) = 0$ forces $X \equiv 0$. Not an eigenvalue.

- $\lambda > 0$: $X = A\cos(\sqrt{\lambda}x) + B\sin(\sqrt{\lambda}x)$. $X(0) = 0$ gives $A = 0$; $X(l) = 0$ with $B \neq 0$ gives $\sin(\sqrt{\lambda}l) = 0$, so $\sqrt{\lambda}l = n\pi$.

$$\lambda_n = \frac{n^2\pi^2}{l^2}, \qquad X_n(x) = \sin\left(\frac{n\pi x}{l}\right), \qquad n = 1, 2, \ldots \tag{118}$$

The eigenvalues are distinct, so orthogonality is automatic; only $\|X_n\|^2 = l/2$ still has to be computed. (Choksi rules out $\lambda < 0$ directly on p. 461.)

12.6 The Four Standard Boundary Conditions

Condition	Eigenvalues	Eigenfunctions	$\|X_n\|^2$
Dirichlet on $[0,l]$: $X(0) = 0 = X(l)$	$\frac{n^2\pi^2}{l^2}$, $n \geq 1$	$\sin\frac{n\pi x}{l}$	$\frac{l}{2}$
Neumann on $[0,l]$: $X'(0) = 0 = X'(l)$	$\frac{n^2\pi^2}{l^2}$, $n \geq 0$	$\cos\frac{n\pi x}{l}$	l; $\frac{l}{2}$
Periodic on $[-l,l]$	$\frac{n^2\pi^2}{l^2}$, $n \geq 0$	1; $\cos\frac{n\pi x}{l}$, $\sin\frac{n\pi x}{l}$	$2l$; l
Mixed on $[0,l]$: $X(0) = 0 = X'(l)$	$\frac{(n+\frac{1}{2})^2\pi^2}{l^2}$, $n \geq 0$	$\sin\frac{(n+\frac{1}{2})\pi x}{l}$	$\frac{l}{2}$

The first three give the sine, cosine, and full Fourier series.

Remark 39.

- *Neumann: $\lambda = 0$ is an eigenvalue (constant eigenfunction), the $\frac{1}{2}a_0$ term.*
- *Periodic: each λ_n, $n \geq 1$, is* degenerate*, with two eigenfunctions* cos *and* sin*. The theorem says nothing about two eigenfunctions with the same eigenvalue. These two happen to be orthogonal (their product is odd), but* sin *and* cos + sin *span the same eigenspace and are not. Within a degenerate eigenspace choose an orthogonal pair (Gram–Schmidt).*
- *Mixed: a new series, reused in the pointwise convergence proof.*

Example (In class, Choksi 11.10(b))**.** *$X'' + \lambda X = 0$, $X(0) = 0$, $X'(l) = 0$. Green's first identity (117) still applies (at l the factor X' vanishes, at 0 the factor X), so $\lambda \geq 0$. For $\lambda = 0$, $X = A + Bx$ with $A = 0$ and $B = 0$: not an eigenvalue. For $\lambda > 0$, $X = B\sin(\sqrt{\lambda}x)$ and $X'(l) = B\sqrt{\lambda}\cos(\sqrt{\lambda}l) = 0$, so $\sqrt{\lambda}l = (n + \frac{1}{2})\pi$, the mixed row of the table.*

12.7 General Fourier Series

Fix a symmetric boundary condition with mutually orthogonal eigenfunctions $\{X_n\}$ and eigenvalues $\{\lambda_n\}$ (a degenerate eigenvalue listed once per eigenfunction of an orthogonal pair). If $\phi = \sum A_n X_n$ can be integrated term by term against each X_n, then

$$A_n = \frac{(\phi, X_n)}{\|X_n\|^2} = \frac{\int_a^b \phi X_n \, dx}{\int_a^b X_n^2 \, dx}. \tag{119}$$

Definition 31 (11.3.1, general Fourier series)**.** *With A_n from (119), $\sum_n A_n X_n$ is the general Fourier series of ϕ. For Dirichlet, Neumann, and periodic conditions it is a* classical *Fourier series.*

Remark 40 (Why this basis)**.** *$\mathcal{A}$ is diagonal: formally, $\mathcal{A}\big(\sum A_n X_n\big) = \sum \lambda_n A_n X_n$. Two previews (formal; term-by-term differentiation needs Choksi 11.6):*

- *$-u'' = f$ on $(0,l)$, $u(0) = u(l) = 0$: with $f = \sum f_n X_n$ and $u = \sum u_n X_n$, each mode gives $\lambda_n u_n = f_n$, so $u_n = f_n/\lambda_n$. Inverting the operator is division, possible since no Dirichlet eigenvalue is 0.*
- *$u_t = \alpha u_{xx}$ with Dirichlet ends: $u = \sum u_n(t) X_n(x)$ gives $u_n' = -\alpha\lambda_n u_n$, so $u_n(t) = u_n(0)e^{-\alpha\lambda_n t}$. High frequencies decay fastest (Section 14).*

13 Fourier Series C: Convergence

October 19, 2026

Source: Choksi 11.4, 11.5, 11.7; Strauss 5.4. Question 1: does the Fourier series of ϕ converge to ϕ? The one idea: **the Fourier partial sum is an orthogonal projection, so it is the closest approximation.**

13.1 Modes of Convergence

Definition 32 (Modes of convergence)**.** *For f_N, f on (a,b):*

- *pointwise if $f_N(x) \to f(x)$ for every $x \in (a,b)$ (11.5.1);*
- *uniformly on $[a,b]$ if $\sup_{x\in[a,b]} |f_N(x) - f(x)| \to 0$;*
- *in L^2 (mean-square) if $\int_a^b |f_N - f|^2\,dx \to 0$, i.e. $\|f_N - f\| \to 0$ (11.4.1).*

Definition 33 (L^2)**.** *$L^2((a,b))$ is the set of f with $\int_a^b |f|^2\,dx < \infty$.*

Example. *Every piecewise continuous function on (a,b) is in L^2. So is the unbounded $|x|^{-1/4}$ on $(-1,1)$, but not $|x|^{-1/2}$.*

Remark 41. *On a bounded interval, uniform implies both others, since $\int_a^b |f_N - f|^2 \le (b-a)\sup|f_N - f|^2$. No other implication holds: on $(0,1)$, $x^N \to 0$ pointwise and in L^2 ($\int_0^1 x^{2N} = \frac{1}{2N+1}$), but $\sup x^N = 1$.*

Proposition 18 (Cauchy–Schwarz)**.** *For $f, g \in L^2((a,b))$, $|(f,g)| \le \|f\|\,\|g\|$. (Same proof as for vectors, Assignment 3.) So Fourier coefficients of any $\phi \in L^2$ are defined.*

13.2 Best Approximation and Bessel's Inequality

Let $\{X_n\}$ be *any* orthogonal family of nonzero functions, $\phi \in L^2$, $A_n = (\phi, X_n)/\|X_n\|^2$, and $S_N = \sum_{n=1}^N A_n X_n$.

Theorem 11 (Least squares). *Among all $\sum_{n=1}^{N} c_n X_n$, the closest to ϕ in L^2 is S_N ($c_n = A_n$), and*

$$\|\phi - S_N\|^2 = \|\phi\|^2 - \sum_{n=1}^{N} A_n^2 \|X_n\|^2. \tag{120}$$

Proof. Expand; orthogonality removes the cross terms, then complete the square in each c_n:

$$\left\|\phi - \sum c_n X_n\right\|^2 = \|\phi\|^2 - 2\sum c_n(\phi, X_n) + \sum c_n^2 \|X_n\|^2 \tag{121}$$

$$= \sum \|X_n\|^2 (c_n - A_n)^2 + \|\phi\|^2 - \sum A_n^2 \|X_n\|^2. \tag{122}$$

The c_n appear only in the first sum, which is ≥ 0 and $= 0$ exactly when $c_n = A_n$. □

Remark 42 (The picture). *$(\phi - S_N, X_n) = (\phi, X_n) - A_n\|X_n\|^2 = 0$ for $n \leq N$, so S_N is the orthogonal projection onto $\operatorname{span}\{X_1, \ldots, X_N\}$ and (120) is Pythagoras. Adding X_{N+1} leaves the earlier coefficients alone.*

Theorem 12 (11.3, Bessel's inequality). *For $\phi \in L^2((a,b))$ and any orthogonal family,*

$$\sum_{n=1}^{\infty} A_n^2 \int_a^b X_n^2 \, dx \leq \int_a^b \phi^2 \, dx. \tag{123}$$

Proof. The left side of (120) is ≥ 0; let $N \to \infty$. □

It holds for every orthogonal family, complete or not (with a gap for the sines missing $\sin(\pi x/l)$).

Theorem 13 (11.5, Riemann–Lebesgue lemma). *If $\phi \in L^2((a,b))$, its classical Fourier coefficients tend to 0.*

Proof. For the classical series $\|X_n\|^2$ is the same for all $n \geq 1$, so Bessel gives $\sum A_n^2 < \infty$, and terms of a convergent series tend to 0. □

Remark 43. *Compare $\sin(nx) \to 0$ in Section 10, whose proof needed a derivative of the probe. Bessel needs none: any square-integrable ϕ is a valid probe.*

13.3 Parseval's Equality and Completeness

Now let $\{X_n\}$ be the eigenfunctions of a symmetric boundary condition. By (120), $\|\phi - S_N\| \to 0$ exactly when the gap in Bessel closes.

Proposition 19. *The general Fourier series of $\phi \in L^2$ converges to ϕ in L^2 if and only if*

$$\sum_{n=1}^{\infty} A_n^2 \int_a^b X_n^2 \, dx = \int_a^b \phi^2 \, dx \qquad \textit{(Parseval's equality).} \tag{124}$$

Definition 34 (Complete)**.** *An orthogonal family is complete in $L^2((a,b))$ if Parseval's equality holds for every $\phi \in L^2((a,b))$. This is "the miracle" made precise.*

Theorem 14 (11.2 and 11.4, L^2 convergence)**.** *For any symmetric boundary condition and any $\phi \in L^2((a,b))$, Parseval's equality holds and the general Fourier series converges to ϕ in L^2. The eigenfunctions are complete.*

Not proved: it needs the spectral theorem for symmetric operators, beyond the course.

Example (Parseval computes sums)**.** *$\phi \equiv 1$ on $(0,\pi)$, sine series: $\|X_n\|^2 = \pi/2$, $A_n = \frac{4}{n\pi}$ for n odd. Then*

$$\sum_{n \text{ odd}} \left(\frac{4}{n\pi}\right)^2 \frac{\pi}{2} = \int_0^\pi 1\,dx = \pi, \qquad \text{so} \qquad 1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \cdots = \frac{\pi^2}{8}. \tag{125}$$

Example (In class, Choksi 11.11(b))**.** *$\phi(x) = x$ on $(-\pi,\pi)$ is odd, so $a_n = 0$ and $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)\,dx = \frac{2(-1)^{n+1}}{n}$. With $\|\sin(nx)\|^2 = \pi$, Parseval gives*

$$\sum_{n=1}^{\infty} \frac{4}{n^2}\,\pi = \int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^3}{3}, \qquad \text{so} \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{126}$$

13.4 The Dirichlet Kernel

L^2 convergence says nothing about a given point. Take the full series on $(-\pi,\pi)$:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \phi(y)\cos(ny)\,dy, \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \phi(y)\sin(ny)\,dy, \tag{127}$$

$$S_N(x) = \frac{1}{2}a_0 + \sum_{n=1}^{N} \big[a_n\cos(nx) + b_n\sin(nx)\big]. \tag{128}$$

Substitute the coefficients (finite sum, so it goes inside the integral) and use $\cos(ny)\cos(nx) + \sin(ny)\sin(nx) = \cos\big(n(x-y)\big)$:

$$S_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \left[1 + 2\sum_{n=1}^{N} \cos\big(n(x-y)\big)\right] \phi(y)\,dy. \tag{129}$$

Definition 35 (Dirichlet kernel)**.** *For $N \geq 1$, the 2π-periodic function $K_N(x) = 1 + 2\sum_{n=1}^{N}\cos(nx)$.*

So the partial sum is a convolution:

$$S_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} K_N(x-y)\phi(y)\,dy. \tag{130}$$

Proposition 20 (Properties of K_N)**.**

(1) $K_N(0) = 2N+1$*, and* K_N *is even.*

(2) $\frac{1}{2\pi}\int_{-\pi}^{\pi} K_N(x)\,dx = 1$ *(every cosine integrates to* 0*).*

(3) For x *not a multiple of* 2π*,*

$$K_N(x) = \frac{\sin\left((N+\frac{1}{2})x\right)}{\sin(x/2)}. \tag{131}$$

Proof of (3). $K_N(x) = \sum_{n=-N}^{N} e^{inx}$, a geometric sum with ratio $e^{ix} \neq 1$. Sum it and multiply top and bottom by $e^{-ix/2}$:

$$K_N(x) = e^{-iNx}\frac{1-e^{i(2N+1)x}}{1-e^{ix}} = \frac{e^{-i(N+\frac{1}{2})x} - e^{i(N+\frac{1}{2})x}}{e^{-ix/2} - e^{ix/2}} = \frac{\sin\left((N+\frac{1}{2})x\right)}{\sin(x/2)}. \tag{132}$$

□

Remark 44. *For* $S_N(x) \to \phi(x)$ *we need* $\frac{1}{2\pi}K_N(x-y)$ *to act like* δ_x*. Unit mass and a peak hold, but* K_N *is not nonnegative and its oscillations away from* 0 *do not shrink. So the proof of Theorem 5.1 does not apply, and continuity of* ϕ *will not be enough.*

13.5 Pointwise Convergence

Theorem 15 (11.6, pointwise convergence of the full series)**.** *Suppose* ϕ *and* ϕ' *are piecewise continuous on* $(-\pi,\pi)$ *and* $\phi((-\pi)+)$*,* $\phi'((-\pi)+)$*,* $\phi(\pi-)$*,* $\phi'(\pi-)$ *exist. Extend* ϕ *periodically to* $\mathbb{R}$*. Then for every* x*,*

$$S_N(x) \to \frac{\phi(x+) + \phi(x-)}{2}, \tag{133}$$

and $S_N(x) \to \phi(x)$ *wherever the periodic extension is continuous.*

Proof when the periodic extension is C^1*.* Fix x. In (130) substitute $y = x+\theta$; since K_N is even and everything is 2π-periodic, $S_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} K_N(\theta)\phi(x+\theta)\,d\theta$. By property (2), $\phi(x) = \frac{1}{2\pi}\int K_N(\theta)\phi(x)\,d\theta$. Subtract and use (131):

$$S_N(x) - \phi(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \psi(\theta)\sin\left((N+\tfrac{1}{2})\theta\right)d\theta, \qquad \psi(\theta) = \frac{\phi(x+\theta)-\phi(x)}{\sin(\theta/2)}. \tag{134}$$

An orthogonal family. $X_N(\theta) = \sin\left((N+\frac{1}{2})\theta\right)$, $N = 0,1,2,\dots$, are the mixed eigenfunctions ($X(0) = 0 = X'(\pi)$), hence orthogonal on $(0,\pi)$. Each is odd, so they are also orthogonal on $(-\pi,\pi)$, with $\|X_N\|^2 = \pi$. So

$$S_N(x) - \phi(x) = \frac{1}{2\pi}(\psi, X_N) = \frac{1}{2}B_N, \qquad B_N = \frac{(\psi, X_N)}{\|X_N\|^2}. \tag{135}$$

Bessel finishes it, if $\psi \in L^2$. Bessel gives $\sum_N B_N^2 \pi \le \|\psi\|^2$, so $B_N \to 0$ and $S_N(x) \to \phi(x)$.

ψ is bounded. $\sin(\theta/2)$ vanishes on $[-\pi, \pi]$ only at $\theta = 0$. By the mean value theorem $|\phi(x+\theta) - \phi(x)| \le M|\theta|$ with $M = \max|\phi'|$, and $|\sin(\theta/2)| \ge |\theta|/\pi$ for $|\theta| \le \pi$ (concavity of sin on $[0, \pi/2]$). So $|\psi| \le \pi M$ and $\psi \in L^2$. □

Remark 45. *The case with jumps splits the integral at $\theta = 0$ and uses $\phi(x+)$ on $(0, \pi)$ and $\phi(x-)$ on $(-\pi, 0)$ (Choksi p. 476). The derivative was used only to bound ψ near 0. It cannot be dropped entirely: some continuous functions have Fourier series diverging at a point.*

Corollary 15.1.

- *For $\phi \in C_c^\infty((-\pi, \pi))$ and $x = 0$: $\frac{1}{2\pi} \int K_N \phi = S_N(0) \to \phi(0)$, so $K_N \to 2\pi\delta_0$ in the sense of distributions on $(-\pi, \pi)$ (the preview of Section 10).*
- *The convergence theorem of Section 11 follows for the sine and cosine series by applying Theorem 11.6 to the odd or even extension (Assignment 3).*

13.6 Uniform Convergence and the Gibbs Phenomenon

Theorem 16 (11.9, uniform convergence)**.** *If ϕ is continuous and piecewise C^1 on $[-\pi, \pi]$ with $\phi(-\pi) = \phi(\pi)$, its full Fourier series converges absolutely and uniformly to the periodic extension of ϕ on $\mathbb{R}$.*

Remark 46. *$\phi(-\pi) = \phi(\pi)$ says the periodic extension has no jumps. It is needed: each S_N is continuous, and a uniform limit of continuous functions is continuous. The square wave is a counterexample.*

Remark 47 (Gibbs phenomenon)**.** *Near a jump the partial sums overshoot. For the square wave ± 1, the first maximum of S_N right of 0 tends to $\frac{2}{\pi} \int_0^\pi \frac{\sin x}{x}\, dx \approx 1.18$, about 9% of the jump. It does not shrink with N, it only moves toward the jump.*

Remark 48 (Smoothness and decay)**.** *If the periodic extension of ϕ is continuous, the Fourier coefficients of ϕ' are n times those of ϕ (Choksi Theorem 11.7, one integration by parts). Each continuous derivative of the periodic extension buys one more power of $1/n$.*

13.7 Summary: The Three Questions

(1) **Does it work?** In L^2 for any $\phi \in L^2$ and any symmetric condition (stated). Pointwise, to the average at jumps, for ϕ, ϕ' piecewise continuous (proved via the Dirichlet kernel and Bessel). Uniformly for a continuous periodic extension with piecewise continuous derivative; at jumps it cannot, and Gibbs appears.

(2) **What are the coefficients?** $A_n = (\phi, X_n)/\|X_n\|^2$, the orthogonal projection, giving the best L^2 approximation.

(3) **Why sines and cosines?** They are eigenfunctions of the symmetric operator $-\frac{d^2}{dx^2}$ with Dirichlet, Neumann, or periodic conditions. Symmetry gives orthogonality and real eigenvalues.

14 The Diffusion Equation A: Separation of Variables

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Source: Choksi 12.1, 7.1, 7.6; Strauss 4.1, 4.2. The diffusion (heat) equation is

$$u_t = \alpha u_{xx}, \qquad \alpha > 0, \tag{136}$$

where $u(x,t)$ is a temperature or a concentration and α is the diffusivity (Strauss writes k). The initial data is g. In N dimensions, $u_t = \alpha \Delta u$ with $\Delta u = \operatorname{div} \nabla u = u_{x_1 x_1} + \cdots + u_{x_N x_N}$. The one idea: **in the eigenfunction basis the diffusion equation is a decoupled list of ODEs, and mode n decays like $e^{-\alpha \lambda_n t}$. High frequencies die first.**

14.1 The Dirichlet Problem on an Interval

A thin bar on $0 \le x \le l$, insulated along its sides, ends held at temperature 0, initial temperature g:

$$\begin{cases} u_t = \alpha u_{xx}, & 0 < x < l,\ t > 0, \\ u(0,t) = u(l,t) = 0, & t > 0, \\ u(x,0) = g(x), & 0 < x < l. \end{cases} \tag{137}$$

Remark 49. *The boundary conditions are imposed for $t > 0$ only, so g need not vanish at the ends (a bar at temperature 1 plunged into ice water at $t = 0$). They are* homogeneous*: if u, v satisfy them, so does $au + bv$. The PDE is linear too, so superposition applies to both.*

14.2 Separated Solutions

Look for $u(x,t) = X(x)T(t)$. Then $XT' = \alpha X'' T$, and dividing by αXT,

$$\frac{T'(t)}{\alpha T(t)} = \frac{X''(x)}{X(x)}. \tag{138}$$

The left side depends only on t and the right only on x, so both equal a constant $-\lambda$:

$$X'' + \lambda X = 0, \qquad T' = -\alpha \lambda T. \tag{139}$$

The boundary conditions give $X(0)T(t) = 0 = X(l)T(t)$, so for a nonzero solution $X(0) = 0 = X(l)$. This is the Dirichlet eigenvalue problem (Section 12):

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \qquad X_n(x) = \sin\left(\frac{n\pi x}{l}\right), \qquad n = 1, 2, \ldots \tag{140}$$

Then $T_n(t) = e^{-\alpha\lambda_n t}$, and

$$u_n(x,t) = \sin\left(\frac{n\pi x}{l}\right) \exp\left(-\frac{\alpha n^2\pi^2 t}{l^2}\right) \tag{141}$$

solves the PDE and boundary conditions: a fixed shape decaying in place. Mode n has $n-1$ interior zeros and decays n^2 times faster than mode 1.

Remark 50. *λ is not a choice: the boundary conditions force it to be an eigenvalue. All eigenvalues are real and positive, so there are no growing modes.*

14.3 Superposition and the Initial Condition

Take an infinite sum of the u_n:

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\alpha n^2\pi^2 t/l^2}. \tag{142}$$

At $t = 0$ this is $\sum b_n \sin(n\pi x/l)$, so to match g take the Fourier sine coefficients:

$$b_n = \frac{2}{l}\int_0^l g(x)\sin\left(\frac{n\pi x}{l}\right) dx. \tag{143}$$

Method 6 (Separation of variables, Choksi p. 498)**.** *(i) Find separated solutions.*

(ii) Impose the boundary conditions to get an eigenvalue problem.

(iii) Solve for T_n.

(iv) Superpose.

(v) Choose the coefficients by projection.

Steps (i) to (iii) find the eigenfunctions; (iv) and (v) find the coefficients.

Example (Hot bar plunged into ice water, Strauss 4.1 Exercise 2)**.** *$g \equiv 1$. From Section 11, $b_n = \frac{4}{n\pi}$ for n odd and 0 for n even, so*

$$u(x,t) = \frac{4}{\pi}\sum_{n \text{ odd}} \frac{1}{n}\sin\left(\frac{n\pi x}{l}\right) e^{-\alpha n^2\pi^2 t/l^2}. \tag{144}$$

At $t = 0$ the series converges slowly, like $\sum 1/n$, to a function that jumps at the ends. For any $t > 0$ the factors $e^{-\alpha n^2\pi^2 t/l^2}$ make it converge faster than any geometric series. Once αt is comparable to l^2, only the first term matters:

$$u(x,t) \approx \frac{4}{\pi}\sin\left(\frac{\pi x}{l}\right) e^{-\alpha\pi^2 t/l^2}. \tag{145}$$

The ratio of the third term to the first is $\frac{1}{3}e^{-8\alpha\pi^2 t/l^2}$, about 10^{-4} at $t = l^2/(\alpha\pi^2)$.

Remark 51 (Time scale)**.** *The bar cools on the time scale $l^2/(\alpha\pi^2)$. Doubling the length multiplies the cooling time by four: time scales like length squared.*

Example (In class)**.** *$l = \pi$, $\alpha = 1$, $g(x) = \sin x + 3\sin 5x$. No integrals are needed: g is already a finite sine series, so $b_1 = 1$, $b_5 = 3$, and*

$$u(x,t) = e^{-t}\sin x + 3e^{-25t}\sin 5x. \tag{146}$$

At $t = 0.1$: $u \approx 0.90\sin x + 0.25\sin 5x$, the wiggles are already much smaller. At $t = 1$: $3e^{-25} \approx 4\times 10^{-11}$, so $u \approx e^{-1}\sin x$, a single hump.

Example (Easy exam question)**.** *$\alpha = 1$, $l = \pi$, $g(x) = 3\sin x + 10\sin 7x$ on $(0,\pi)$. Again g is already a sine series, so $b_1 = 3$, $b_7 = 10$, and*

$$u(x,t) = 3e^{-t}\sin x + 10e^{-7^2 t}\sin 7x. \tag{147}$$

A harder exam question would give a general g.[2]

14.4 Neumann Boundary Conditions

Insulate the ends as well:

$$\begin{cases} u_t = \alpha u_{xx}, & 0 < x < l,\ t > 0, \\ u_x(0,t) = u_x(l,t) = 0, & t > 0, \\ u(x,0) = g(x), & 0 < x < l. \end{cases} \tag{148}$$

Separation now gives $X'' + \lambda X = 0$, $X'(0) = 0 = X'(l)$: $\lambda_n = (n\pi/l)^2$, $X_n = \cos(n\pi x/l)$, $n \geq 0$. The new feature is $\lambda_0 = 0$ with constant X_0 and constant T_0:

$$u(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) e^{-\alpha n^2\pi^2 t/l^2}, \qquad a_n = \frac{2}{l}\int_0^l g(x)\cos\left(\frac{n\pi x}{l}\right) dx. \tag{149}$$

As $t \to \infty$ every mode with $n \geq 1$ dies, and

$$u(x,t) \to \frac{1}{2}a_0 = \frac{1}{l}\int_0^l g(x)\,dx. \tag{150}$$

The bar equalizes to its average initial temperature.

Proposition 21 (Conservation of heat)**.** *If u is a smooth solution of (148), then $\int_0^l u(x,t)\,dx$ is constant in t.*

Proof.

$$\frac{d}{dt}\int_0^l u\,dx = \int_0^l \alpha u_{xx}\,dx = \alpha\big(u_x(l,t) - u_x(0,t)\big) = 0. \qquad \square \tag{151}$$

So the only constant the solution can approach is the average. Under Dirichlet conditions the boundary term does not vanish and heat leaves through the ends.

[2] For a hard version, compute the Fourier sine coefficients of g with (143), then put them into (142), as in the hot bar example.

14.5 Where the Equation Comes From

Measure heat so that the heat in $[a,b]$ is $\int_a^b u\,dx$, and let $F(x,t)$ be the *flux*: the rate at which heat crosses x from left to right.

Conservation. Heat in $[a,b]$ changes only by flow through the endpoints:

$$\frac{d}{dt}\int_a^b u\,dx = F(a,t) - F(b,t) = -\int_a^b F_x\,dx. \tag{152}$$

This holds for every $[a,b]$, so $u_t = -F_x$.

Fourier's law (Fick's law). Heat flows from hot to cold at a rate proportional to the slope:

$$F = -\alpha u_x. \tag{153}$$

Together, $u_t = \alpha u_{xx}$. In N dimensions the divergence theorem gives $u_t = -\operatorname{div}\vec{F}$ and $\vec{F} = -\alpha\nabla u$, so $u_t = \alpha\Delta u$. (Choksi Exercise 7.1: how α depends on density, specific heat, and conductivity.) The physics is in (153): read u_x as minus the flux.

14.6 Boundary Conditions and Heat Flow

Read each condition at the left end $x = 0$:

- **Dirichlet**, $u(0,t) = T_0$: the end touches a reservoir so large that its temperature never changes. Homogeneous case $T_0 = 0$.
- **Neumann**, $u_x(0,t) = 0$: the flux $F(0,t) = -\alpha u_x(0,t)$ is 0, the end is insulated.
- **Robin**: the end touches a medium at T_0, and heat passes at a rate proportional to the temperature difference (Newton's law of cooling). The heat entering at $x = 0$ is $F(0,t)$, so for some $h > 0$

$$-\alpha u_x(0,t) = h\big(T_0 - u(0,t)\big), \qquad \text{i.e.} \qquad u_x(0,t) = \frac{h}{\alpha}\big(u(0,t) - T_0\big). \tag{154}$$

  At the right end the heat leaving is $F(l,t)$, giving $-\alpha u_x(l,t) = h\big(u(l,t) - T_0\big)$. As $h \to 0$ this becomes Neumann; as $h \to \infty$, Dirichlet.

Remark 52 (A sign to check)**.** *If $T_0 > u(0,t)$, heat enters, $F(0,t) > 0$, so $u_x(0,t) < 0$: the temperature decreases going into the bar. Choksi (7.6.2, p. 300) writes $u_x(0,t) = c_1(T_0 - u(0,t))$ and states $u_x(0,t) > 0$, the opposite sign. The form above agrees with Strauss 4.3 and Choksi Exercises 12.5 and 12.7.*

Steady states (Choksi (7.30) to (7.32), 12.3.1). A steady state $v(x) = \lim_{t\to\infty} u(x,t)$ satisfies $v'' = 0$, so v is linear, and the boundary conditions pick the line:

Boundary conditions	Steady state $v(x)$
$u(0,t) = T_0,\ u(l,t) = T_1$	$T_0 + (T_1 - T_0)\,x/l$
$u(0,t) = T_0,\ u_x(l,t) = 0$	T_0
$u_x(0,t) = 0 = u_x(l,t)$	$\frac{1}{l}\int_0^l g\,dx$

For Neumann conditions every constant satisfies $v'' = 0$ and the boundary conditions; conservation of heat picks the constant.

Remark 53 (Inhomogeneous Dirichlet conditions)**.** *Separation needs homogeneous conditions (two solutions with $u(0,t) = T_0$ sum to $2T_0$). Subtract the steady state: if u solves the first row and v is its steady state, $w = u - v$ solves (137) with data $g - v$. So*

$$u(x,t) = v(x) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\alpha n^2\pi^2 t/l^2}, \tag{155}$$

with b_n the sine coefficients of $g - v$. The solution approaches the line at rate $\alpha\pi^2/l^2$. The second row uses the mixed eigenfunctions $\sin\left((n+\frac{1}{2})\pi x/l\right)$ (Choksi 12.3.2, Exercise 12.3).

14.7 The Solution Operator and the Kernel (Optional)

For a symmetric matrix A with orthonormal eigenvectors $\vec{v}_n$ and eigenvalues λ_n, the system $\vec{x}\,' = -\alpha A\vec{x}$, $\vec{x}(0) = \vec{x}_0$, has solution

$$\vec{x}(t) = \sum_n e^{-\alpha\lambda_n t}(\vec{x}_0 \cdot \vec{v}_n)\,\vec{v}_n. \tag{156}$$

The coordinates decouple and each decays on its own. Formula (142) is this with $A = -\frac{d^2}{dx^2}$ and Dirichlet conditions: with orthonormal $e_n = \sqrt{2/l}\,\sin(n\pi x/l)$ and $c_n(t) = (u(\cdot,t), e_n)$,

$$c_n'(t) = -\alpha\lambda_n c_n(t), \qquad c_n(0) = (g, e_n). \tag{157}$$

Definition 36 (Solution operator)**.** *$S(t)g = u(\cdot,t)$. It multiplies the nth coordinate by $e^{-\alpha\lambda_n t}$.*

Proposition 22.

(1) $S(t)S(s) = S(t+s)$: running for time s then t is running for $t+s$.

(2) $\|S(t)g\| \le e^{-\alpha\lambda_1 t}\|g\|$, by Parseval. The L^2 norm decreases.

The kernel. Put (143) into (142) and exchange sum and integral (allowed for $t > 0$ since the terms decay so fast):

$$u(x,t) = \int_0^l K(x,y,t)\,g(y)\,dy, \qquad K(x,y,t) = \frac{2}{l}\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right)\sin\left(\frac{n\pi y}{l}\right) e^{-\alpha n^2\pi^2 t/l^2}. \tag{158}$$

The solution at x is a weighted sum of the data with weight $K(x,\cdot,t)$ (Choksi calls it Φ in (12.7); here Φ is reserved for the heat kernel on the line).

Remark 54 (Shape of K)**.**

- *For small t it is a narrow bump at $y = x$, matching the Gaussian heat kernel on the line away from the ends: over a short time only data near x matters.*
- *It is never negative and vanishes at $y = 0, l$: data at the ends, held at 0, contribute nothing.*
- *Its total weight $\int_0^l K\,dy$ decreases in t, faster near the ends. The missing weight is heat lost through the ends, so u is a weighted average with total weight at most 1 and shrinks toward 0.*

14.8 Summary

- Separation of variables reduces the diffusion equation on an interval to the eigenvalue problem for $-\frac{d^2}{dx^2}$ with the given boundary condition; the coefficients are Fourier coefficients of the data.
- Mode n decays like $e^{-\alpha\lambda_n t}$. So the solution is C^∞ for every $t > 0$, settles onto the first mode at rate $\alpha\lambda_1$, and the problem cannot be run backward.
- In the eigenbasis the solution operator is diagonal, as for $\vec{x}\,' = -\alpha A\vec{x}$.
- Boundary conditions are statements about the flux $F = -\alpha u_x$: fixed temperature, zero flux, or flux proportional to a temperature difference.
- Still open: is (142) the only solution? Answered in Lecture B without the formula.

A Appendix

B Solutions

Each solution links back to its problem. Problems are placed in the section where their topic belongs.

B.1 Assignment 1

Problem 5 (A1 Q1, vector calculus), p. 7

(1) Let $\vec{x} = (x, y, z)$, so $f = x^2 + y^2 + z^2$. Then

$$\nabla f = (2x, 2y, 2z), \qquad \nabla f(1,0,0) = (2,0,0),$$

$$D_{\vec{v}} f(1,0,0) = \nabla f(1,0,0) \cdot \vec{v} = (2,0,0) \cdot \tfrac{1}{\sqrt{2}}(1,1,0) = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

The level sets $f = c$ are spheres of radius $\sqrt{c}$. In the plane ($z = 0$), $f = x^2 + y^2$ has circular contours (e.g. $c = 1, 4, 9$), and $\nabla f = 2\vec{x}$ points radially outward, perpendicular to each circle, longer farther out.

(2) Linear approximation: $f(\vec{x} + h\vec{v}) \approx f(\vec{x}) + \nabla f(\vec{x}) \cdot h\vec{v}$, so

$$D_{\vec{v}} f(\vec{x}) = \lim_{h \to 0} \frac{f(\vec{x} + h\vec{v}) - f(\vec{x})}{h} = \nabla f(\vec{x}) \cdot \vec{v}.$$

By Cauchy–Schwarz, $D_{\vec{v}} f(\vec{x}) \le \|\nabla f(\vec{x})\| \, \|\vec{v}\| = \|\nabla f(\vec{x})\|$. Since $\nabla f(\vec{x}) \neq \vec{0}$, set $\vec{v}^* = \nabla f(\vec{x}) / \|\nabla f(\vec{x})\|$; then $\nabla f \cdot \vec{v}^* = \|\nabla f\|^2 / \|\nabla f\| = \|\nabla f\|$, so the bound is attained. Uniqueness: if $\nabla f \cdot \vec{v} = \|\nabla f\|$, dividing by $\|\nabla f\|$ gives $\vec{v} \cdot \vec{v}^* = 1$, and

$$\|\vec{v} - \vec{v}^*\|^2 = \|\vec{v}\|^2 - 2\vec{v} \cdot \vec{v}^* + \|\vec{v}^*\|^2 = 1 - 2 + 1 = 0,$$

so $\vec{v} = \vec{v}^*$.

(3)

- $f = x^2 + 2y^2 + 3z^2$: $f_x = 2x$, $f_y = 4y$, $f_z = 6z$ vanish only at the origin, and $H = \operatorname{diag}(2, 4, 6)$ with leading minors $\Delta_1 = 2$, $\Delta_2 = 8$, $\Delta_3 = 48$, all positive. Local minimum.
- $f = x^2 + y^2 - z^2$: critical point $(0,0,0)$, $H = \operatorname{diag}(2, 2, -2)$, $\Delta_1 = 2$, $\Delta_2 = 4$, $\Delta_3 = -8$. Saddle point.
- $f = xy$: $f_x = y$, $f_y = x$, critical point $(0,0)$, $H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\Delta_1 = 0$, $\Delta_2 = -1$. Eigenvalues: $\det(H - \lambda I) = \lambda^2 - 1 = 0$, so $\lambda = \pm 1$. Mixed signs, so a saddle point.[3]

[3] Viewed as a function on $\mathbb{R}^3$, the Hessian also has the eigenvalue 0 from x_3; the mixed signs ± 1 still give a saddle.

Problem 1 (A1 Q2, Choksi 1.1), p. 2

For each: (I) classification, (II) context, (III) domain and auxiliary conditions.

(b) Inviscid Burgers, $u_t + uu_x = 0$. (I) First order, quasilinear. (II) Models 1D gas dynamics: u is the fluid velocity, x position, t time; no parameters. (III) $x \in \mathbb{R}$, $t > 0$, $u(x,0) = g(x)$. Shocks form, so weak (entropy) solutions are needed.

(d) Hamilton–Jacobi, $u_t + H(\nabla u, x) = 0$. (I) First order, fully nonlinear. (II) In classical mechanics u is the action, x the generalized coordinates, H the Hamiltonian, with ∇u as momentum. In optimal control u is the value function. (III) $x \in \mathbb{R}^N$, $t > 0$, $u(x,0) = g(x)$.

(e) KdV, $u_t + u_{xxx} - 6uu_x = 0$. (I) Third order, semilinear. (II) Models long waves in shallow water: u is the surface elevation, x position along the channel, t time. The coefficients come from scaling; u_{xxx} is dispersion and $-6uu_x$ is the nonlinearity. (III) $x \in \mathbb{R}$, $t > 0$, $u(x,0) = g(x)$, u decaying as $|x| \to \infty$.

(i) Black–Scholes, $u_t + \frac{1}{2}\sigma^2 S^2 u_{SS} + rSu_S - ru = 0$. (I) Second order, linear. (II) u is the price of a European option, S the stock price, t time, σ the volatility, r the risk-free interest rate. (III) $S > 0$, $0 \le t \le T$, terminal condition $u(S,T) = \max(S - K, 0)$ for a call with strike K, boundary condition $u(0,t) = 0$.

(j) Monge–Ampère, $\det(D^2 u) = f$. (I) Second order, fully nonlinear. (II) Prescribes the Gaussian curvature of the graph of u. In optimal transport, ∇u maps one distribution onto another and f is the ratio of the two densities. (III) x in a bounded convex domain Ω, $u = g$ on $\partial\Omega$, $f > 0$, u convex.

(o) Fisher–KPP, $u_t = au_{xx} + bu(1 - u)$. (I) Second order, semilinear. (II) Models the spread of an advantageous gene in a population: u is the gene frequency (a normalized density), x position, t time, a the diffusion coefficient, b the growth rate. (III) $x \in \mathbb{R}$, $t > 0$, $u(x,0) = g(x)$ with $0 \le g \le 1$.

Problem 2 (A1 Q2, Choksi 1.2), p. 4

Assume u is smooth enough and F, G, H are arbitrary functions.

(a) u does not depend on x: $u = F(y,z)$.

(b) Integrating in y gives $u_x = g(x,z)$; integrating in x then gives $u = F(x,z) + G(y,z)$.

(c) Integrating in z gives $u_{xy} = h(x,y)$. Applying (b) with z as a parameter gives $u = F(x,y) + G(x,z) + H(y,z)$.

Problem 4 (A1 Q2, Choksi 1.6), p. 5

(a) Existence for every $\vec{b}$ means A is surjective, i.e. rank $A = N$. Uniqueness means $\ker A = \{\vec{0}\}$. For a square matrix these are equivalent, and both hold iff A is invertible ($\det A \neq 0$).

(b) Stability means small changes in $\vec{b}$ produce small changes in $\vec{x}$: there is a constant C with

$$A\vec{x}_1 = \vec{b}_1,\ A\vec{x}_2 = \vec{b}_2 \implies \|\vec{x}_1 - \vec{x}_2\| \leq C\|\vec{b}_1 - \vec{b}_2\|.$$

Since $\vec{x}_1 - \vec{x}_2 = A^{-1}(\vec{b}_1 - \vec{b}_2)$, $C = \|A^{-1}\|$. For A symmetric positive definite, $\|A^{-1}\| = 1/\lambda_{\min}$, so $\|\vec{x}_1 - \vec{x}_2\| \leq \frac{1}{\lambda_{\min}}\|\vec{b}_1 - \vec{b}_2\|$. The smaller $\lambda_{\min}$, the more a perturbation in $\vec{b}$ is amplified, so the less stable the problem.

Problem 3 (A1 Q2, Choksi 1.10), p. 5

Let $u_n(x,t) = \frac{1}{n}e^{n^4 t}\sin(nx)$. Then $\partial_t u_n = n^3 e^{n^4 t}\sin(nx) = \partial_x^4 u_n$, so u_n solves the PDE with initial data $g_n(x) = \frac{1}{n}\sin(nx)$. The zero data $g \equiv 0$ gives the solution $u \equiv 0$, and the data g_n get close to 0 since $|g_n(x)| \leq \frac{1}{n} \to 0$ for all x. But for any fixed $t > 0$, taking $x = \frac{\pi}{2n}$ gives $u_n(\frac{\pi}{2n}, t) = \frac{e^{n^4 t}}{n} \to \infty$. Small initial data produce large solutions: the solution does not depend continuously on the data, so the problem is not well-posed.

Problem 6 (A1 Q3, Choksi 2.3), p. 11

u is constant along lines with direction (a,b). The points $(1,2)$ and $(3,6)$ are on such a line, so $(a,b) \parallel (3-1, 6-2) = (2,4)$. Hence $b = 2a$ and $\dfrac{a}{b} = \dfrac{1}{2}$.

Problem 8 (A1 Q3, Choksi 2.4), p. 12

The characteristics satisfy $\frac{dy}{dx} = \frac{b}{a} = y^2$, which gives $x + \frac{1}{y} = c$. So u is constant along each curve $y = \frac{1}{c-x}$, which has two branches: $y > 0$ for $x < c$ and $y < 0$ for $x > c$ (they are different characteristics). The data:

point	c	branch	u
$(3,2)$	$7/2$	$y > 0$	7
$(4,2)$	$9/2$	$y > 0$	-6
$(8,1)$	9	$y > 0$	-2
$(6,-1)$	5	$y < 0$	3
$(6,-2)$	$11/2$	$y < 0$	0
$(10,7)$	$71/7$	$y > 0$	8
$(15,-4)$	$59/4$	$y < 0$	10

- $(\frac{5}{2}, \frac{1}{2})$: $c = \frac{5}{2} + 2 = \frac{9}{2}$, $y > 0$, same characteristic as $(4,2)$, so $u = -6$.
- $(8, -\frac{2}{5})$: $c = 8 - \frac{5}{2} = \frac{11}{2}$, $y < 0$, same characteristic as $(6,-2)$, so $u = 0$.

- $(\frac{9}{2}, 1)$: $c = \frac{9}{2} + 1 = \frac{11}{2}$, but $y > 0$, and the only data point with $c = \frac{11}{2}$ is $(6, -2)$, on the $y < 0$ branch. Cannot be determined.

Problem 7 (A1 Q3, Choksi 2.5), p. 11

$\frac{dy}{dx} = -\frac{3}{4}$, so $3x + 4y = C$ on characteristics and $u = f(3x + 4y)$. On $y = 5x$: $f(3x + 20x) = f(23x) = e^x$, so $f(s) = e^{s/23}$ and

$$u(x, y) = e^{(3x+4y)/23}.$$

Problem 13 (A1 Q3, Choksi 2.6), p. 15

Characteristics. $\frac{dx}{dt} = x$, so $x = x_0 e^t$ and $x_0 = xe^{-t}$. Also $\frac{du}{dt} = 0$, so $u(x, t) = g(x_0) = \frac{1}{x_0^2+1}$, giving

$$u(x, t) = \frac{1}{x^2 e^{-2t} + 1}.$$

Backward flow. $\phi_t(x_0) = x_0 e^t$, so $\phi_{-t}(x) = xe^{-t}$ and $u(x, t) = g(\phi_{-t}(x)) = \frac{1}{x^2 e^{-2t}+1}$. Both methods agree.

Profiles. Every point moves away from 0 at speed $|x|$: the blip keeps height 1 at $x = 0$ and spreads out, getting wider as $t = 0, 1, 2, 3, 4$ increases (the opposite of the narrowing in the $v = -x$ example).

Problem 11 (A1 Q4, Choksi 2.9), p. 14

Characteristic system: $\frac{dx}{ds} = y$, $\frac{dy}{ds} = 3$, $\frac{du}{ds} = -u$, with $x(0) = x_0$, $y(0) = 0$, $u(0) = \sin x_0$. Then $y = 3s$, $\frac{dx}{ds} = 3s$ gives $x = x_0 + \frac{3s^2}{2}$, and $u = \sin(x_0)e^{-s}$. Inverting, $s = \frac{y}{3}$ and $x_0 = x - \frac{3}{2} \cdot \frac{y^2}{9} = x - \frac{y^2}{6}$, so

$$u(x, y) = e^{-y/3} \sin\left(x - \frac{y^2}{6}\right).$$

Check: $u_x = e^{-y/3} \cos(x - \frac{y^2}{6})$, $u_y = -\frac{y}{3} e^{-y/3} \cos(x - \frac{y^2}{6}) - \frac{1}{3} e^{-y/3} \sin(x - \frac{y^2}{6})$, so $yu_x + 3u_y = -e^{-y/3} \sin(x - \frac{y^2}{6}) = -u$, and $u(x, 0) = \sin x$.

Problem 12 (A1 Q4, Choksi 2.10), p. 14

$\frac{dy}{dx} = x$, so $y = \frac{x^2}{2} + C$ and $u = f(y - \frac{x^2}{2})$. Then $u(x, 0) = f(-\frac{x^2}{2}) = e^x$. But $f(-\frac{x^2}{2})$ is even in x and e^x is not: for any $a \neq 0$, $e^a = f(-\frac{a^2}{2}) = e^{-a}$ forces $a = 0$, a contradiction. Any neighbourhood of the origin contains $(\pm a, 0)$ for small $a > 0$, so no solution exists near the origin.

Picture. The characteristics are upward parabolas $y = \frac{x^2}{2} + C$. Those with $C < 0$ cross the data line $y = 0$ twice, at $x = \pm\sqrt{-2C}$. E.g. the $C = -1$ parabola forces $u(-\sqrt{2}, 0) = u(\sqrt{2}, 0)$, but the data gives $e^{-\sqrt{2}} \neq e^{\sqrt{2}}$. The data line meets these characteristics twice, not exactly once.

Problem 9 (A1 Q4, Choksi 2.21), p. 12

$y-x^2-2x = C$ on each curve, so $u = f(y-x^2-2x)$. Then $u_x = -(2x+2)f'$ and $u_y = f'$, so

$$u_x + (2x+2)u_y = 0.$$

Check: the characteristic slope $\frac{dy}{dx} = 2x+2$ matches $\frac{d}{dx}(x^2+2x+C)$.

B.2 Assignment 2

Problem 15 (A2 Q1, Choksi 5.2), p. 21

Step 1: f'. Since $f = 0$ on $(-\infty, 0)$, $\langle f', \phi\rangle = -\int_{-\infty}^{\infty} f\phi'\,dx = -\int_0^{\infty} x\phi'(x)\,dx$. Integrate by parts with $u = x$, $dv = \phi' dx$:

$$-\int_0^{\infty} x\phi'\,dx = -\left(\big[x\phi(x)\big]_0^{\infty} - \int_0^{\infty}\phi\,dx\right).$$

At $x = 0$ the factor x is 0, and for large x, $\phi = 0$. So $\langle f', \phi\rangle = \int_0^{\infty}\phi\,dx = \int_{-\infty}^{\infty} H(x)\phi(x)\,dx$, i.e. $f' = H$.

Step 2: f''. $\langle f'', \phi\rangle = -\langle H, \phi'\rangle = -\int_0^{\infty}\phi'\,dx = -\left(\lim_{R\to\infty}\phi(R) - \phi(0)\right) = \phi(0) = \langle \delta_0, \phi\rangle$. So $f'' = \delta_0$.

Problem 16 (A2 Q2, Choksi 5.3), p. 21

Step 1. For $x \in (2m-1, 2m+1]$, $f(x) = |x-2m|$, so $f(k) = 0$ for even k and $f(k) = 1$ for odd k. On $(2m, 2m+1)$, $f = x - 2m$ and $f' = 1$; on $(2m-1, 2m)$, $f = 2m - x$ and $f' = -1$. Let g be this square wave, with value g_k on $(k, k+1)$: $g_k = +1$ for even k, -1 for odd k.

Step 2: f'. Choose an integer N with $\operatorname{supp}\phi \subset [-N, N]$. Then $\langle f', \phi\rangle = -\int_{-N}^{N} f\phi' = -\sum_{k=-N}^{N-1}\int_k^{k+1} f\phi'$. On each $[k, k+1]$, f is C^1 and continuous at the endpoints, so

$$\int_k^{k+1} f\phi' = f(k+1)\phi(k+1) - f(k)\phi(k) - \int_k^{k+1} g\phi.$$

The boundary terms telescope: $\sum_{k=-N}^{N-1}\big(f(k+1)\phi(k+1) - f(k)\phi(k)\big) = f(N)\phi(N) - f(-N)\phi(-N) = 0$ since $\phi(\pm N) = 0$. So $\langle f', \phi\rangle = \int g\phi$, i.e. $f' = g$.

Step 3: f''. $\langle f'', \phi\rangle = -\int g\phi' = -\sum_k g_k\int_k^{k+1}\phi' = -\sum_k g_k\big(\phi(k+1) - \phi(k)\big)$. Reindexing with $j = k+1$,

$$\langle f'', \phi\rangle = \sum_j g_j\phi(j) - \sum_j g_{j-1}\phi(j) = \sum_j (g_j - g_{j-1})\phi(j).$$

For even j the jump is $1 - (-1) = 2$, for odd j it is $-1 - 1 = -2$. Hence

$$f'' = \sum_{m\in\mathbb{Z}}\big(2\delta_{2m} - 2\delta_{2m+1}\big).$$

Only finitely many terms are nonzero for each ϕ, so this is a well-defined distribution.

Sketches. f: a triangle wave between 0 (at even integers) and 1 (at odd integers). f': a square wave, $+1$ on $(2m, 2m+1)$, -1 on $(2m-1, 2m)$. f'': upward spikes of weight 2 at even integers, downward spikes of weight -2 at odd integers.

Problem 17 (A2 Q3, Choksi 5.4), p. 21

Take $f(x) = \frac{x^2}{2}$ for $x \geq 0$ and $f(x) = 0$ for $x < 0$.

Step 1. $\langle f', \phi\rangle = -\int_0^\infty \frac{x^2}{2}\phi'\,dx = -\left[\frac{x^2}{2}\phi\right]_0^\infty + \int_0^\infty x\phi\,dx = \int_0^\infty x\phi\,dx$, so $f' = \max(x, 0)$.

Step 2. By Problem 15, $f'' = H$, which is also a function.

Step 3. Again by Problem 15, $\langle f''', \phi\rangle = -\int_0^\infty \phi' = \phi(0)$, so $f''' = \delta_0$.

Problem 18 (A2 Q4, Choksi 5.9; not submitted, solution added)

Integrate the function part and add a jump of size 1 at $x = 2$:

$$f(x) = \frac{x^3}{3} + 2x^2 + H(x-2).$$

$\frac{x^3}{3} + 2x^2$ is C^1, so its derivative is $x^2 + 4x$ classically, and $H(x-2)' = \delta_2$ by the Heaviside computation translated to 2. (Any constant may be added.)

Problem 19 (A2 Q4, several jumps; not submitted, solution added)

Fix ϕ vanishing outside $[-R, R]$ with $-R < x_1$, $x_k < R$, and set $x_0 = -R$, $x_{k+1} = R$. Split and integrate by parts on each piece $[x_j, x_{j+1}]$, where f and f' extend continuously from inside:

$$\langle f', \phi\rangle = -\sum_{j=0}^{k}\int_{x_j}^{x_{j+1}} f\phi' = \sum_{j=0}^{k}\int_{x_j}^{x_{j+1}} g\phi - \sum_{j=0}^{k}\Big(f(x_{j+1}^-)\phi(x_{j+1}) - f(x_j^+)\phi(x_j)\Big).$$

The first sum is $\int g\phi$. In the second, the end terms vanish since $\phi(\pm R) = 0$, and collecting the two terms at each interior x_j gives $\sum_{j=1}^{k}\big(f(x_j^+) - f(x_j^-)\big)\phi(x_j)$. So $f' = g + \sum_j \big(f(x_j^+) - f(x_j^-)\big)\delta_{x_j}$. For $k = 1$ this is the theorem of Section 9; Problem 16 is the case of jumps of f' at every integer.

Problem 14 (A2 Q5, Choksi 5.11), p. 19

Let $\langle T, \phi\rangle = \sum_{n=1}^{\infty}\phi(n)$.

Step 1: finite. If $\operatorname{supp}\phi \subset [-R, R]$, then $\phi(n) = 0$ for $n > R$, so the sum has at most $\lfloor R\rfloor$ nonzero terms.

Step 2: linear. The sums are finite, so $\sum(a\phi + b\psi)(n) = a\sum\phi(n) + b\sum\psi(n)$.

Step 3: continuous. If $\phi_k \to \phi$ in $C_c^\infty(\mathbb{R})$, all supports lie in one $[-R, R]$ and $\phi_k \to \phi$ uniformly, so

$$|\langle T, \phi_k\rangle - \langle T, \phi\rangle| \le \sum_{n \le R} |\phi_k(n) - \phi(n)| \le R \sup |\phi_k - \phi| \to 0.$$

So T is a distribution.

Step 4: antiderivative. Let $f(x) = 0$ for $x < 1$ and $f(x) = n$ for $n \le x < n+1$, $n \ge 1$: the staircase $f = \sum_{n\ge 1} H(x-n)$. With $N = \lfloor R \rfloor$, and $\phi' = 0$ beyond R,

$$\langle f', \phi\rangle = -\int f\phi' = -\sum_{n=1}^{N} n \int_n^{n+1} \phi' = -\sum_{n=1}^{N} n\big(\phi(n+1) - \phi(n)\big).$$

Reindexing the first sum: $\langle f', \phi\rangle = \sum_{n=1}^{N} n\phi(n) - \sum_{m=2}^{N+1}(m-1)\phi(m) = \sum_{n=1}^{N}\phi(n) - N\phi(N+1)$. Since $N+1 > R$, $\phi(N+1) = 0$, so $\langle f', \phi\rangle = \sum_{n\ge 1}\phi(n) = \langle T, \phi\rangle$.

Problem 25 (A2 Q6, Choksi 5.6), p. 25

Step 1. $\int f_n = \frac{n}{2}\cdot\frac{2}{n} = 1$.

Step 2. Using Step 1, $\langle f_n, \phi\rangle - \phi(0) = \frac{n}{2}\int_{-1/n}^{1/n}\big(\phi(x) - \phi(0)\big)\,dx$.

Step 3. By the mean value theorem, $|\phi(x) - \phi(0)| \le |x|\,\|\phi'\|_\infty \le \frac{\|\phi'\|_\infty}{n}$ for $|x| \le \frac{1}{n}$. So

$$|\langle f_n, \phi\rangle - \phi(0)| \le \frac{n}{2}\cdot\frac{2}{n}\cdot\frac{\|\phi'\|_\infty}{n} = \frac{\|\phi'\|_\infty}{n}.$$

Step 4. Given $\epsilon > 0$, take $N > \|\phi'\|_\infty/\epsilon$. Then $|\langle f_n, \phi\rangle - \phi(0)| < \epsilon$ for $n > N$, so $f_n \to \delta_0$.

Problem 27 (A2 Q7, Choksi 5.12), p. 25

(a) f_1 is -1 on $[0, \frac{1}{2}]$ and 1 on $(\frac{1}{2}, 1)$; f_2 and f_4 are the same square wave compressed to periods $\frac{1}{2}$ and $\frac{1}{4}$. Yes, $f_n \to 0$ in the sense of distributions. Each f_n oscillates between -1 and 1 with period $\frac{1}{n}$, and its integral over each period $I_k = [\frac{k}{n}, \frac{k+1}{n}]$ is $\frac{1}{n}(-\frac{1}{2} + \frac{1}{2}) = 0$. Since ϕ is continuous, it is nearly constant on each short period, so $\int_{I_k} f_n\phi \approx \phi(\frac{k}{n})\int_{I_k} f_n = 0$. Summing over the periods meeting $\operatorname{supp}\phi \subset [-R, R]$ gives

$$|\langle f_n, \phi\rangle| \le \left(\frac{2R}{n} + \frac{2}{n^2}\right)\|\phi'\|_\infty \to 0.$$

(b) With values -1 and 5, $f_n \to 2$. The average of f over one period is $\frac{-1+5}{2} = 2$ (so $\int_{I_k} f_n = \frac{1}{n}(-\frac{1}{2} + \frac{5}{2}) = \frac{2}{n} = \int_{I_k} 2$). Then $f_n - 2$ takes the values -3 and 3 and has average 0 over each period, and the argument of (a) applied to $f_n - 2$ gives $\langle f_n - 2, \phi\rangle \to 0$, i.e. $f_n \to 2$: the distributional limit is the mean value.

Problem 26 (A2 Q8, Choksi 5.14), p. 25

Lemma (continuum version of Theorem 5.3). Let g be integrable with $\int g = 1$ and $g_\sigma(x) = \frac{1}{\sigma} g(\frac{x}{\sigma})$. Then $g_\sigma \to \delta_0$ as $\sigma \to 0^+$.

Step 1. With $y = x/\sigma$, $dx = \sigma\, dy$: $\langle g_\sigma, \phi\rangle = \int g(y)\phi(\sigma y)\, dy$.

Step 2. Since $\int g = 1$: $\langle g_\sigma, \phi\rangle - \phi(0) = \int g(y)\big(\phi(\sigma y) - \phi(0)\big)\, dy$.

Step 3. Given $\epsilon > 0$, choose R with $\int_{|y|>R} |g| < \epsilon$ (possible since g is integrable). For $|y| > R$, $|g(y)(\phi(\sigma y) - \phi(0))| \le 2\|\phi\|_\infty |g(y)|$, so that piece is at most $2\|\phi\|_\infty \epsilon$. For $|y| \le R$, the mean value theorem gives $|\phi(\sigma y) - \phi(0)| \le \sigma|y|\, \|\phi'\|_\infty \le \sigma R \|\phi'\|_\infty$, so that piece is at most $\sigma R \|\phi'\|_\infty \int |g|$.

Step 4. Adding the two pieces and letting $\sigma \to 0$: $\limsup_{\sigma\to 0} |\langle g_\sigma, \phi\rangle - \phi(0)| \le 2\|\phi\|_\infty \epsilon$. Since ϵ is arbitrary, the limit is 0, so $g_\sigma \to \delta_0$.

Application. Take $g(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$, integrable with $\int g = 1$. Then $g_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}$. With $\sigma = \frac{1}{\sqrt{n}}$, so that $\sigma^2 = \sigma_n$, this is exactly $f_n(x) = \frac{1}{\sqrt{2\pi\sigma_n}} e^{-x^2/(2\sigma_n)}$. Since $\sigma = \frac{1}{\sqrt{n}} \to 0$, the lemma gives $f_n \to \delta_0$.

Problem 28 (A2 Q9, Choksi 5.17; not submitted, solution added)

Write $f = \frac{A}{\ell} + h$, where $h = f - \frac{A}{\ell}$ is ℓ-periodic with integral 0 over each period. The constant gives $\int \frac{A}{\ell}\phi$ for every n. For h, let $G(x) = \int_0^x h$; it is ℓ-periodic (zero mean) and bounded by $M = \int_0^\ell |h|$. Since $\frac{d}{dx}\big[\frac{1}{n}G(nx)\big] = h(nx)$, integrating by parts as in Theorem 5.4,

$$\left|\int h(nx)\phi(x)\, dx\right| = \left|\frac{1}{n}\int G(nx)\phi'(x)\, dx\right| \le \frac{M}{n}\int |\phi'|\, dx \to 0.$$

So $f_n \to \frac{A}{\ell}$ in the sense of distributions. Problem 27 is the case $\ell = 1$, $A = 0$ and $A = 2$.

Problem 29 (A2 Q10, Choksi 5.18), p. 26

Step 1. Let $h_n = n\,\mathbf{1}_{[-1/n,1/n]}$. Then

$$\langle h_n', \phi\rangle = -n\int_{-1/n}^{1/n} \phi' = n\phi(-\tfrac{1}{n}) - n\phi(\tfrac{1}{n}) = \langle F_n, \phi\rangle,$$

so $F_n = h_n'$.

Step 2. $h_n = 2f_n$ with f_n the hats, so by Problem 25, $h_n \to 2\delta_0$.

Step 3. By Proposition 5.5.1, $F_n = h_n' \to 2\delta_0'$, where $\langle 2\delta_0', \phi\rangle = -2\phi'(0)$.

Check: $n\big(\phi(-\frac{1}{n}) - \phi(\frac{1}{n})\big) = -2\,\dfrac{\phi(\frac{1}{n}) - \phi(-\frac{1}{n})}{2/n} \to -2\phi'(0)$.

Problem 30 (A2 Q11, Choksi 5.19; not submitted, solution added)

Differentiate an approximation to the identity. With the spikes $g_n = n - n^2|x|$ on $|x| < \frac{1}{n}$, let

$$f_n = g_n' = \begin{cases} n^2, & -\frac{1}{n} < x < 0, \\ -n^2, & 0 < x < \frac{1}{n}, \\ 0, & \text{otherwise.} \end{cases}$$

g_n is continuous, so f_n is also its distributional derivative. By Theorem 5.1, $g_n \to \delta_0$, so by Proposition 5.5.1, $f_n \to \delta_0'$. Sketch: a positive block of height n^2 on $(-\frac{1}{n}, 0)$ next to a negative block of height $-n^2$ on $(0, \frac{1}{n})$, taller and narrower as $n = 1, 5, 10, 20$ (a dipole, as in Problem 29).

Problem 20 (A2 Q12), p. 22

Step 1: uniqueness and symmetry. If u_1, u_2 both solve the problem, $w = u_1 - u_2$ satisfies $w'' = 0$ on $[-1, 1]$, so $w = ax + b$, and $w(-1) = -a + b = 0$, $w(1) = a + b = 0$ give $a = b = 0$. Symmetry: $v(x) = u(-x)$ has $v''(x) = u''(-x) = f_\epsilon(-x) = f_\epsilon(x)$ and $v(\pm 1) = u(\mp 1) = 0$, so by uniqueness $v = u$: u is even. Differentiating $u(-x) = u(x)$ gives $-u'(-x) = u'(x)$, and $x = 0$ gives $u'(0) = 0$.

Step 2: solve for $x \geq 0$. u' is continuous at $\pm\frac{\epsilon}{2}$ because $u'(x) = u'(0) + \int_0^x f_\epsilon$, and u is continuous because it is differentiable, so u and u' can be matched at $x = \frac{\epsilon}{2}$.

- On $[0, \frac{\epsilon}{2}]$: $u'' = \frac{1}{\epsilon}$ and $u'(0) = 0$, so $u' = \frac{x}{\epsilon}$ and $u = \frac{x^2}{2\epsilon} + b$.
- On $[\frac{\epsilon}{2}, 1]$: $u'' = 0$ and $u' = u'(\frac{\epsilon}{2}) = \frac{1}{2}$. With $u(1) = 0$, $u = \frac{1}{2}(x - 1)$.
- Continuity of u at $\frac{\epsilon}{2}$: $\frac{\epsilon}{8} + b = \frac{\epsilon}{4} - \frac{1}{2}$, so $b = \frac{\epsilon}{8} - \frac{1}{2}$.

Step 3: extend evenly.

$$u_\epsilon(x) = \begin{cases} \dfrac{x^2}{2\epsilon} + \dfrac{\epsilon}{8} - \dfrac{1}{2}, & |x| \leq \frac{\epsilon}{2}, \\ \dfrac{|x| - 1}{2}, & \frac{\epsilon}{2} \leq |x| \leq 1. \end{cases}$$

Step 4: limit. Let $u_0 = \frac{|x|-1}{2}$. For $|x| \geq \frac{\epsilon}{2}$, $u_\epsilon = u_0$. For $|x| \leq \frac{\epsilon}{2}$,

$$u_\epsilon - u_0 = \frac{x^2 - \epsilon|x| + \frac{\epsilon^2}{4}}{2\epsilon} = \frac{(|x| - \frac{\epsilon}{2})^2}{2\epsilon} \in \left[0, \frac{\epsilon}{8}\right].$$

So $\sup |u_\epsilon - u_0| \leq \frac{\epsilon}{8}$, and for $\phi \in C_c^\infty((-1, 1))$, $|\langle u_\epsilon, \phi\rangle - \langle u_0, \phi\rangle| \leq \frac{\epsilon}{8} \int |\phi| \to 0$.

Note: the limit is $u_0 = \Phi - \frac{1}{2}$, not Φ exactly. The boundary conditions $u(\pm 1) = 0$ force the constant $-\frac{1}{2}$. It differs from Φ by a constant and has the same second derivative δ_0.

Problem 21 (A2 Q13), p. 22

Use $\langle T'', \phi\rangle = \langle T, \phi''\rangle$.
Φ_+: integrate by parts with $u = x$, $dv = \phi'' dx$:

$$\langle \Phi_+'', \phi\rangle = \int_0^\infty x\phi''\,dx = \big[x\phi'\big]_0^\infty - \int_0^\infty \phi'\,dx = 0 - \big(0 - \phi(0)\big) = \phi(0).$$

Φ_-: integrate by parts with $u = -x$, $dv = \phi'' dx$:

$$\langle \Phi_-'', \phi\rangle = \int_{-\infty}^0 (-x)\phi''\,dx = \big[-x\phi'\big]_{-\infty}^0 + \int_{-\infty}^0 \phi'\,dx = 0 + \big(\phi(0) - 0\big) = \phi(0).$$

So $\Phi_+'' = \Phi_-'' = \delta_0$.

Problem 22 (A2 Q14), p. 22

The three functions on each half-line:

	Φ	Φ_+	Φ_-
$x \geq 0$	$x/2$	x	0
$x < 0$	$-x/2$	0	$-x$

Subtracting case by case:

- $\Phi_+ - \Phi_- = x$ in both cases ($x - 0$ and $0 - (-x)$).
- $\Phi - \Phi_+ = -\frac{x}{2}$ in both cases ($\frac{x}{2} - x$ and $-\frac{x}{2} - 0$).
- $\Phi - \Phi_- = \frac{x}{2}$ in both cases ($\frac{x}{2} - 0$ and $-\frac{x}{2} + x$).

All three differences are linear. Even: only Φ, since $\Phi(-x) = \frac{1}{2}|-x| = \Phi(x)$; $\Phi_\pm$ are not even, since $\Phi_+(-x) = \Phi_-(x)$. Vanishes for $x < 0$: only Φ_+, since $\Phi_+(x) = \max(x, 0) = 0$ there; the other two are positive there ($\Phi = -\frac{x}{2}$ and $\Phi_- = -x$).

Problem 23 (A2 Q15), p. 22

Step 1. For $y < x$, $|x - y| = x - y$; for $y > x$, $|x - y| = y - x$. So

$$u(x) = \frac{1}{2}\int_{-\infty}^x (x - y)f(y)\,dy + \frac{1}{2}\int_x^\infty (y - x)f(y)\,dy.$$

Step 2. Let $A(x) = \int_{-\infty}^x f$, $B(x) = \int_{-\infty}^x yf(y)\,dy$, $c_0 = \int f$, $c_1 = \int yf$, all finite since f has compact support. Then $\int_x^\infty f = c_0 - A$ and $\int_x^\infty yf = c_1 - B$, so

$$2u = xA - B + (c_1 - B) - x(c_0 - A) = 2xA - 2B + c_1 - c_0 x.$$

Step 3. Since f and yf are continuous, the Fundamental Theorem of Calculus gives $A' = f$ and $B' = xf(x)$. So

$$2u' = 2A + 2xf - 2xf - c_0 = 2A - c_0, \qquad 2u'' = 2f.$$

Hence $u'' = f$ classically.

Problem 24 (A2 Q16), p. 22

Step 1. If $\psi = \chi'$ with $\chi \in C_c^\infty(\mathbb{R})$, then for large R, $\int \psi = \chi(R) - \chi(-R) = 0$.

Step 2. Conversely, suppose $\int \psi = 0$ and $\operatorname{supp}\psi \subset [-R, R]$. Define $\chi(x) = \int_{-\infty}^{x} \psi$. Then $\chi' = \psi$, so χ is smooth. For $x < -R$, $\chi(x) = 0$; for $x > R$, $\chi(x) = \int \psi = 0$. So $\chi \in C_c^\infty(\mathbb{R})$.

Step 3. Fix $\phi_0 \in C_c^\infty(\mathbb{R})$ with $\int \phi_0 = 1$. For any ϕ, the function $\psi = \phi - (\int \phi)\phi_0$ satisfies $\int \psi = 0$, so by Step 2, $\psi = \chi'$ for some $\chi \in C_c^\infty(\mathbb{R})$.

Step 4. $\langle G, \psi\rangle = \langle G, \chi'\rangle = -\langle G', \chi\rangle = 0$.

Step 5. By linearity, $\langle G, \phi\rangle = \langle G, \psi\rangle + (\int \phi)\langle G, \phi_0\rangle = c\int \phi$ with $c := \langle G, \phi_0\rangle$. So G is the constant function c.

Step 6. Let $T'' = \delta_0$ and $W = T - \frac{1}{2}|x|$. Then $W'' = \delta_0 - \delta_0 = 0$. Since $(W')' = 0$, Step 5 gives $W' = a$ for a constant a. Also the derivative of the function ax is a, because $-\int ax\phi' = \int a\phi$. So $(W - ax)' = 0$, and Step 5 gives $W - ax = b$. Therefore $T = \frac{1}{2}|x| + ax + b$.

C Useful Links